

A REPORT ON

“Smart Metering of Electricity theft Detection and Power Monitoring”

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF
THE DEGREE

OF

**BACHELOR OF ENGINEERING (COMPUTER
ENGINEERING)**

SUBMITTED BY

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2024-25



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PROJECT APPROVAL SHEET

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Abstract

The increasing demand for electricity and the associated challenges of power theft and inefficiency have necessitated innovative solutions in energy management. This report explores the implementation of smart metering technologies for the detection of electricity theft and enhanced power monitoring. Smart meters utilize advanced communication technologies and real-time data analytics to provide accurate consumption readings, enabling utilities to identify anomalies indicative of theft. By integrating these systems with predictive algorithms and machine learning techniques, this approach not only facilitates immediate detection but also aids in predicting potential theft hotspots. Additionally, the report discusses the broader implications of smart metering on energy efficiency, consumer behavior, and grid management. The findings highlight the potential of smart metering as a pivotal tool in enhancing the reliability and sustainability of electrical distribution systems, ultimately contributing to reduced losses and improved service delivery.

Key words: Keywords: Electricity Theft, Power Monitoring, IoT, Power Theft Detection, NodeMCU.

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CHAPTER 1

INTRODUCTION

The rapid advancement in technology has significantly transformed the energy sector, with smart metering emerging as a cornerstone of modern power management. Traditional electricity meters are being replaced by smart meters, which provide real-time, accurate data on energy consumption and enable two-way communication between consumers and utility providers. This shift is essential for optimizing power distribution, reducing operational costs, and enhancing the efficiency of energy systems.

One of the most pressing issues faced by utility companies worldwide is electricity theft, which results in substantial financial losses and disrupts the balance of power supply and demand. Smart metering systems, combined with advanced data analytics and machine learning techniques, offer a robust solution for detecting and preventing electricity theft. By analyzing consumption patterns, these systems can identify irregularities that signal potential tampering or unauthorized use, allowing for proactive measures to combat energy losses.

Power monitoring is another critical aspect of smart metering technology. It involves the continuous observation of electrical parameters to ensure the stability and reliability of the power grid. Through precise measurements of voltage, current, and power quality, utility providers can quickly respond to issues, maintain optimal grid performance, and reduce downtime caused by faults or outages.

Despite its benefits, the adoption of smart metering and electricity theft detection technologies faces several challenges, including concerns about data privacy, infrastructure costs, and resistance from consumers. However, advancements in the Internet of Things (IoT), artificial intelligence (AI), and blockchain technology are paving the way for smarter, more secure energy solutions.

This paper explores the principles and technologies behind smart metering, electricity theft detection, and power monitoring, examining their role in modern energy management. It discusses the benefits and challenges associated with these technologies and highlights emerging trends that are set to redefine the future of energy systems.

1. MOTIVATION OF THE PROJECT

The adoption of smart metering and electricity theft detection technologies is driven by the need to enhance power grid efficiency, reduce revenue losses, and support sustainability goals. Electricity theft poses significant financial losses for utilities, which can be mitigated through real-time data analytics and anomaly detection. Smart meters empower consumers with detailed insights into their energy consumption, promoting cost savings and environmentally friendly habits. Additionally, these technologies facilitate the transition to smart grids by improving demand management, supporting the integration of renewable energy, and leveraging AI and IoT for better decision-making. Embracing these innovations is crucial for creating a more reliable, secure, and sustainable energy ecosystem.

2. Problem Definition

Traditional power distribution systems face significant challenges, including electricity theft, inefficient energy monitoring, and poor power quality management. Electricity theft leads to substantial revenue losses for utilities, while the lack of real-time consumption data hinders grid optimization and decision-making. Consumers have limited control over their energy usage due to inadequate insights into their consumption patterns. Additionally, integrating renewable energy into the grid remains a challenge, and concerns about data privacy and security persist. Addressing these issues requires innovative solutions that enhance monitoring, reduce losses, and support a more efficient, secure, and sustainable energy infrastructure.

1.3 OBJECTIVE STATEMENTS

1. Develop and implement smart metering systems to enable real-time monitoring and accurate measurement of electricity consumption.
2. Create data-driven methods for detecting and preventing electricity theft to reduce non-technical losses and increase revenue protection.
3. Enhance power quality monitoring to identify and address issues like voltage fluctuations and harmonic distortions, ensuring reliable power supply.
4. Empower consumers with detailed insights into their energy usage to promote cost savings and encourage energy-efficient behaviors.
5. Facilitate the seamless integration of renewable energy sources into the power grid for a more sustainable and resilient energy system.
6. Ensure robust data privacy and security measures to protect consumer information and maintain the integrity of smart metering systems.

1.4 Schedule of Project Work

Start Date	End Date	Description	Duration (Days)
19-09-2024	25-09-2024	Topic Selection	6
26-09-2024	05-10-2024	Literature Review	9
06-10-2024	12-10-2024	Existing System Study	6
13-10-2024	20-10-2024	Proposed System Study	7
21-10-2024	25-10-2024	Report Preparation (Initial)	4
26-10-2024	05-11-2024	Requirement Analysis	10
06-11-2024	15-11-2024	System Design	9
16-11-2024	10-12-2024	Hardware Setup & Integration	24
11-12-2024	05-01-2025	Software Implementation	25
06-01-2025	15-01-2025	Module Testing	9
16-01-2025	31-01-2025	System Testing	15
01-02-2025	10-02-2025	Theft Detection Logic Integration	9
11-02-2025	20-02-2025	Dashboard Development	9
21-02-2025	05-03-2025	Final Testing & Debugging	13
06-03-2025	15-03-2025	Result Analysis & Evaluation	9
16-03-2025	31-03-2025	Final Report Writing	15
01-04-2025	10-04-2025	Documentation Formatting	9
11-04-2025	20-04-2025	Plagiarism Check & Corrections	9
21-04-2025	30-04-2025	Project File Compilation	9
01-05-2025	05-05-2025	Final Touch-ups & Submission Prep	5

1.5 Budget of Project

Hardware Components

1. ESP32 Development Boards (2 units)
 1. Consumer side: ₹450
 2. Distributor side: ₹450
 3. Total: ₹900
2. Voltage Sensors
 1. ZMPT101B (2 units × ₹350): ₹700
3. Current Sensors
 1. ACS714 (2 units × ₹300): ₹600
4. Relay Module
 1. Single channel relay (1 unit): ₹80
5. Power Supply Components
 1. Step-down transformers (2 units × ₹200): ₹400
 2. Voltage regulators and capacitors: ₹200
 3. Total: ₹600
6. PCB and Enclosure
 1. PCB fabrication (2 boards): ₹500
 2. Project boxes/enclosures (2 units × ₹200): ₹400
 3. Total: ₹900
7. Miscellaneous Components
 1. Connecting wires: ₹200
 2. Headers and connectors: ₹150
 3. LED indicators: ₹50
 4. Total: ₹400

Development Tools

1. Basic Tools
 1. Soldering iron and accessories: ₹500
 2. Multimeter: ₹600

3. Wire strippers and cutters: ₹200
4. Total: ₹1,300
2. Testing Equipment
 1. Load testing components: ₹400
 2. Test leads and probes: ₹200
 3. Total: ₹600

Total Base Budget: ₹6,080

Extended Budget (Including Contingency)

1. Component Replacements (15% of hardware cost): ₹600
2. Additional Testing Materials: ₹400
3. Backup Components: ₹500
4. Documentation and Printing: ₹400
5. Project Report Binding and Printing: ₹520

Total Extended Budget: ₹8,500

Budget Notes:

1. Prices are approximate and may vary based on market conditions and vendor selection
2. Budget includes reasonable quality components suitable for academic project implementation
3. Extended budget includes contingency for potential component failures during development
4. Costs can be optimized by:
 1. Bulk purchasing components
 2. Using student discounts where available
 3. Reusing available tools and equipment from the college lab
 4. Sharing common tools within project groups

Individual Member Contributions (4 Members)

Per Person Cost Breakdown:

1. Member 1: ₹2,125
 1. Hardware components contribution: ₹1,000
 2. Development tools contribution: ₹475
 3. Documentation & contingency: ₹650
2. Member 2: ₹2,125
 1. Hardware components contribution: ₹1,000
 2. Development tools contribution: ₹475
 3. Documentation & contingency: ₹650
3. Member 3: ₹2,125
 1. Hardware components contribution: ₹1,000
 2. Development tools contribution: ₹475
 3. Documentation & contingency: ₹650
4. Member 4: ₹2,125
 1. Hardware components contribution: ₹1,000
 2. Development tools contribution: ₹475
 3. Documentation & contingency: ₹650

CHAPTER 2

LITERATURE SURVEY

The literature review highlights significant advancements in smart metering, electricity theft detection, and power monitoring technologies. Smart metering systems enhance real-time data acquisition and two-way communication between consumers and utilities, enabling better load management and demand forecasting. Various machine learning techniques, such as Support Vector Machines (SVM) and Neural Networks, have been explored for effectively detecting electricity theft through anomaly detection in consumption patterns.

Research emphasizes the importance of real-time power quality monitoring, utilizing smart meters to identify issues like voltage fluctuations and harmonic distortions. Studies indicate that providing consumers with detailed insights into their energy usage promotes energy efficiency and conservation. Additionally, the integration of renewable energy sources into smart grids is facilitated by smart metering technology, highlighting the need for advanced forecasting and demand response strategies.

However, data privacy and security remain critical concerns, with research proposing frameworks to safeguard consumer information and mitigate cyber threats. Overall, the literature underscores the transformative potential of these technologies in enhancing energy management while identifying ongoing challenges that require further exploration.

2.1 LITERATURE SURVEY

2.1.1 Literature Survey on References

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5. Rhea Prakash, E. A. (2017). Power Theft Identification Using GSM Technology. International Journal of Advanced

2.2 SUMMARY OF LITERATURE REVIEW

The literature highlights advancements in smart metering for real-time data collection, efficient energy distribution, and load management. Machine learning techniques are effectively used for detecting electricity theft through anomaly detection in consumption patterns. Research emphasizes the role of smart meters in monitoring power quality to identify issues like voltage fluctuations. Studies also show that providing consumers with energy usage insights encourages conservation efforts. Integration of renewable energy into smart grids and addressing data privacy and security concerns are also critical areas of focus.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Introduction

The **Software Requirements Specification (SRS)** outlines the software functionality, characteristics, and constraints of the Smart Metering of Electricity Theft Detection and Power Monitoring. It provides an overview of the project, its intended users, and any assumptions or dependencies that must be considered during development.

3.1.1 Project Scope

The scope of this project is to design and implement a **Smart Metering of Electricity Theft Detection and Power Monitoring** using IoT devices to monitor electricity usage on both the consumer and distributor sides. The system will detect discrepancies in power readings, indicating possible electricity theft. The project includes developing the hardware components (IoT modules) as well as software components (API server, mobile application, database).

Objectives:

1. Monitor electricity consumption in real-time on the consumer and distributor sides.
 2. Detect and report electricity theft or unauthorized power usage.
 3. Provide alerts to users via a mobile app and allow them to view their consumption history.
 4. Provide administrators (utility companies) with the ability to manage the system and monitor theft cases.
-

3.1.2 User Classes and Characteristics

The system is designed for the following **user classes**:

1. **Consumers:**
 1. **Characteristics:** General public using the electricity services.
 2. **Skill level:** Varying levels of technical expertise, from novice to experienced mobile users.
 3. **Responsibilities:**
 1. Monitor personal electricity usage in real-time.

2. Receive alerts regarding potential electricity theft.
 3. Access historical electricity consumption data.
 2. **Utility Company Administrators:**
 1. **Characteristics:** Utility company employees who manage electricity distribution and monitor theft detection.
 2. **Skill level:** High-level technical expertise, including the ability to interpret power data and take corrective actions.
 3. **Responsibilities:**
 1. Monitor discrepancies between distributor and consumer-side electricity data.
 2. Investigate potential cases of electricity theft.
 3. Manage user data and system configurations.
 3. **System Developers:**
 1. **Characteristics:** Technicians responsible for maintaining the system, implementing updates, and handling errors.
 2. **Skill level:** High technical expertise in IoT systems, backend API development, and mobile app support.
 3. **Responsibilities:**
 1. Maintain the IoT system's performance and availability.
 2. Fix bugs and manage updates for the mobile app and server infrastructure.
-

3.1.3 Assumptions and Dependencies

The project is built with several **assumptions** and is dependent on certain external factors for its successful operation:

1. **Assumptions:**
 1. The **IoT modules** (both consumer and distributor sides) are correctly installed and continuously powered to provide real-time data.
 2. Consumers will have access to **smartphones** with internet connectivity to use the mobile app for monitoring their electricity consumption.

3. **Reliable network infrastructure** is in place to ensure uninterrupted communication between the IoT modules and the central server.
 4. The electricity theft patterns will be detectable based on clear discrepancies between consumer and distributor power readings.
2. **Dependencies:**
1. The system depends on **stable internet connections** between the IoT devices and the server. In areas with limited or unreliable internet connectivity, the system's real-time data flow may be compromised.
 2. The mobile application is dependent on being compliant with **Android** and **iOS** standards, ensuring compatibility across both platforms.
 3. The project relies on **third-party sensors** (such as ZMPT 101B for voltage and ACS 714 for current) being reliable and accurate to detect theft correctly.
 4. The system depends on **MongoDB** as the database backend and **Node.js** as the server framework. Any changes to these technologies could impact future development and maintenance.

3.2 EXTERNAL INTERFACES REQUIREMENTS

The **external interfaces requirements** define how the system interacts with users, other software, and communication protocols. This includes the user interfaces, software interfaces, and communication interfaces that ensure seamless interactions and data flow.

3.2.1 User Interfaces

The **user interface (UI)** describes how end users (consumers, admins) will interact with the system.

1. **Mobile Application Interface:**

1. **Platform:** The mobile app should be available on both iOS and Android platforms.
2. **Dashboards:** Users should be able to monitor real-time electricity consumption with a visual dashboard (e.g., charts showing daily/weekly/monthly usage).
3. **Notifications:** Users should receive theft alerts, consumption thresholds warnings, and maintenance notifications through push notifications.
4. **Control Options:** Admins and users should have access to settings for configuring alert levels, viewing past usage, and resetting system preferences.
5. **Language Support:** The interface should support multiple languages, depending on the region of deployment.

2. **Web Interface (Optional):**

1. **Browser Compatibility:** The web interface should be accessible via modern browsers like Chrome, Firefox, and Edge.
2. **Admin Features:** Admins should have access to data analytics, theft detection logs, and the ability to manage IoT devices via a centralized control panel.
3. **User Login:** Secure login with username/password or two-factor authentication (2FA) must be available for users and administrators.

3.2.2 Software Interfaces

Software interfaces describe how the system interacts with other software or external applications. This includes APIs, databases, and third-party integrations.

1. **API Interface:**

1. **RESTful API:** The system should expose a REST API to interact with the IoT devices, mobile application, and web interface.
2. **Authentication:** API endpoints should require secure authentication using tokens or OAuth 2.0 to prevent unauthorized access.
3. **Endpoints:** Key API endpoints include:
 1. `/getPowerData`: Retrieves real-time or historical consumption data.
 2. `/sendAlert`: Sends notifications to the mobile application when theft is detected.
 3. `/updateUserSettings`: Allows users or admins to update system settings (e.g., alert thresholds).
4. **JSON Format:** Data exchange should occur in JSON format for compatibility and ease of use.

2. **Database Interface:**

1. **MongoDB/SQL:** The system should interface with a MongoDB or SQL database to store electricity consumption data, user information, and theft detection logs.
2. **Data Access Layer:** A data access layer should exist to allow the API to securely read/write data from the database.
3. **Backup Interface:** Regular backups should be scheduled, and data should be retrievable in case of system failures.

3. **Third-Party Integration:**

1. **Payment Systems:** The system should interface with third-party payment systems for users to pay bills (if applicable).
2. **Energy Provider Systems:** The system may need to integrate with electricity provider systems for billing, consumption tracking, or regulatory reporting.

3.2.3 Communication Interfaces

The **communication interfaces** outline how system components (IoT devices, API server, mobile app) exchange data.

1. **IoT Device Communication:**

1. **Protocol:** Devices should communicate using standard IoT protocols like MQTT or CoAP, which are lightweight and suitable for low-power devices.
2. **Wi-Fi/Bluetooth:** Devices should use Wi-Fi for data transmission to the server. Bluetooth could be used for local configuration and diagnostics.
3. **Data Frequency:** IoT devices should send electricity consumption data every 30 seconds to ensure real-time monitoring.
4. **Encryption:** All communication between IoT devices and the server must be encrypted (e.g., TLS/SSL) to prevent data tampering or eavesdropping.

2. **API Communication:**

1. **HTTPS:** All API communications between the server, mobile app, and IoT devices should occur over HTTPS to ensure secure data transmission.
2. **Timeouts and Retries:** The system must implement timeouts and retries for all API communications to handle network issues gracefully and avoid data loss.

3. **Push Notifications:**

1. **Protocol:** Push notifications to mobile devices (iOS, Android) should use platform-specific services like Firebase Cloud Messaging (FCM) for Android and Apple Push Notification Service (APNs) for iOS.
2. **Real-Time Alerts:** The system must send theft detection alerts to users immediately after identifying discrepancies in power data.
3. **Data Sync:** Notifications should trigger automatic data sync on the user's mobile app for real-time status updates.

These external interface requirements ensure that users have seamless interactions with the system, the system communicates effectively with its components, and third-party service are integrated securely.

3.3 NONFUNCTIONAL REQUIREMENT

Non-functional requirements define the system's quality attributes, including performance, reliability, security, and usability. Here are some common **non-functional requirements** that apply to a Smart Metering Of Electricity Theft Detection and Power Monitoring:

1. Performance

1. **Data Processing Speed:** The system must process and analyse electricity consumption data within 5 seconds of receiving inputs from IoT devices.
2. **Response Time:** The mobile application should retrieve and display real-time data within 2 seconds of the user's request.
3. **Scalability:** The system should handle data from up to 1 million households without performance degradation, allowing for geographic expansion.

2. Reliability

1. **System Uptime:** The system should have an uptime of 99.9%, ensuring continuous monitoring with minimal downtime.
2. **Fault Tolerance:** The system must be able to tolerate hardware malfunctions, such as sensor failure, and continue functioning, providing alerts if the sensors fail.

3. Security

1. **Data Encryption:** All communication between IoT devices, the server, and the mobile application must be encrypted using industry-standard protocols (e.g., TLS for transmission, AES for data storage).
2. **Authentication & Authorization:** Users should be authenticated using secure methods (e.g., two-factor authentication), and only authorized users (e.g., admin) should have access to critical functionalities like data manipulation.
3. **Data Privacy:** Ensure compliance with data privacy regulations such as GDPR by anonymizing user data when necessary and providing clear privacy policies.

4. Usability

1. **Mobile App Usability:** The mobile application interface should follow user-friendly design principles (clear navigation, intuitive icons, responsive design) and support both novice and expert users.
2. **Localization:** The system should support multiple languages based on the regions it operates in, ensuring non-English speaking users can easily interact with the app.

5. Availability

1. **System Availability:** The system should be accessible 24/7, with the ability to notify users in case of scheduled maintenance windows.
2. **Backup:** The system should take regular backups of critical data (e.g., hourly backups of electricity consumption data) to avoid data loss during outages.

6. Maintainability

1. **Modular Design:** The system must be modular to allow easy updates and maintenance. Changes to one part of the system (e.g., the mobile app) should not affect others (e.g., API server).
2. **Error Logging:** Detailed error logs should be maintained, providing administrators with clear insights into failures or malfunctions.

7. Interoperability

1. **Cross-Platform Support:** The mobile application must be compatible with both Android and iOS devices, and the web interface should be accessible via any modern browser.
2. **IoT Standards Compliance:** The system should be built using widely accepted IoT protocols such as MQTT or CoAP to ensure future compatibility with new sensors and devices.

8. Legal & Compliance

1. **Regulatory Compliance:** The system should comply with electricity monitoring and data-sharing regulations in each region where it is deployed.
2. **Auditability:** The system must maintain detailed logs for audit purposes, including user actions, data updates, and theft detection events, ensuring accountability and traceability.

9. Efficiency

1. **Energy Efficiency:** IoT devices should consume minimal power to ensure long-lasting operations without frequent maintenance or battery changes.
2. **Bandwidth Optimization:** Data communication between IoT modules and the server must be optimized to reduce network usage and prevent congestion.

These **non-functional requirements** are essential for ensuring that the system operates efficiently, securely, and reliably, meeting both user and operational expectations.

3.4 SYSTEM REQUIREMENTS

3.4.1 Database Requirements

1. **Type:** NoSQL database (e.g., MongoDB) for handling large datasets, including sensor readings and user details.
2. **Storage:** Must store data for each user including voltage, current, timestamps, and geographical details.
3. **APIs:** A RESTful API should enable communication between IoT devices and the mobile application, storing data on power consumption and potential theft detection.

3.4.2 User Interfaces

1. **Mobile Application:** A user-friendly mobile app (built using React Native) allowing users to monitor power consumption, theft alerts, and history. It should support graphical representations of consumption patterns.
2. **Web Dashboard:** For administrators and utility companies, providing insights into system health, electricity usage, and theft detection patterns.

3.4.3 Hardware Requirements

1. **IoT Modules:**
 1. **Microcontroller:** ESP32 with Wi-Fi for connectivity.
 2. **Sensors:** ZMPT 101B (voltage sensor) and ACS 714 (current sensor).
 3. **Relay Modules:** Single-channel relay to control the power supply on the consumer side.
 4. **Power Supply:** The system must be powered by stable, uninterrupted electricity.

3.4.4 Software Requirements

1. **Programming Languages:**
 1. Microcontroller code: C/C++ with EmonLib for energy calculations.
 2. Server-side: Node.js with Express.js for the API server.

3. Database: MongoDB for storing the user and sensor data.
4. Mobile App: React Native for cross-platform functionality.

2. Frameworks & Libraries:

1. EmonLib for analog-to-digital conversions.
2. Mongoose for database operations.
3. JSON Web Tokens (JWT) for authentication and authorization.

3.4.5 Analysis Models: SDLC Model to be Applied

1. **Agile SDLC:** An Agile approach with iterative development is recommended. This allows continuous testing and integration of IoT components, APIs, and user interfaces, ensuring flexibility in responding to changes and bug fixes during development.

3.5 SYSTEM IMPLEMENTATION PLAN

1. Phase 1: Requirement Analysis and Feasibility Study

1. Gather detailed requirements from utility companies and consumers.
2. Evaluate available hardware and software solutions, including sensors, microcontrollers, and cloud storage options.

2. Phase 2: Design

1. Design the IoT modules (for both distributor and consumer sides), mobile app interfaces, and API server structure.
2. Prepare circuit diagrams for the hardware and architectural design for the software stack.

3. Phase 3: Development

1. **Hardware setup:** Assemble and program the ESP32 microcontrollers with voltage and current sensors.
2. **Software development:** Build the API server, mobile app, and database. Begin implementing the core functionalities such as data collection, theft detection algorithms, and notifications.

4. Phase 4: Testing

1. Perform unit testing on individual IoT modules, ensuring sensors accurately

measure voltage and current.

2. Integration testing for communication between IoT devices, API server, and mobile application.
3. End-to-end testing, simulating electricity theft and monitoring system responses.

5. **Phase 5: Deployment**

1. Set up the system in a controlled environment with actual users for beta testing.
2. Deploy mobile app to Android and iOS platforms and the API server to a cloud service (e.g., AWS).

6. **Phase 6: Maintenance and Updates**

1. Regular updates to improve theft detection algorithms.
2. Ongoing monitoring of system performance and scalability, addressing hardware malfunctions, and software errors.

4.0 SYSTEM DESIGN

The system design for Smart Metering in Electricity Theft Detection and Power Monitoring is crafted to illustrate the interaction and flow of data between various components, utilizing system architecture, (DFDs), and Language (UML) diagrams. The system architecture provides an overarching view of core elements such as smart meters, data collection units (DCUs), a central server, and the communication network, detailing how data is collected, transmitted, and processed for both theft detection and power monitoring. DFDs break down these processes further by illustrating data sources, key functions, and storage points, while mapping the flow of data from the consumer's meter to the central server for analysis. To dive deeper into system interactions and behaviors, UML diagrams like use case, class, sequence, and activity diagrams offer specific insights into functionalities and processes. Together, these diagrams form a comprehensive framework for visualizing how the system efficiently detects electricity theft and enables robust power usage monitoring, ensuring clear data flow and organized control throughout the network.

4.1 System Architecture

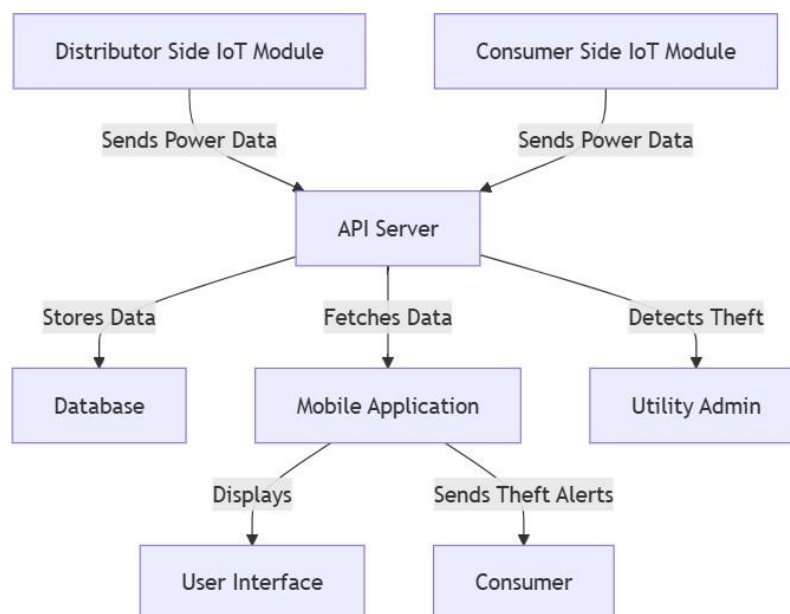


Fig.1.Architecture

The system architecture includes three key components:

1. **Distributor Side IoT Module:**

1. Measures the power sent from the distributor to the consumer.
2. Contains sensors such as ZMPT 101B (voltage) and ACS 714 (current), interfaced with the ESP32 microcontroller.
3. Sends data (voltage, current) to the server via Wi-Fi.

2. **Consumer Side IoT Module:**

1. Measures the consumer's actual electricity usage.
2. Compares data with the distributor-side readings to detect discrepancies (theft).
3. Data is processed using the ESP32 and sent to the server.

3. **API Server and Database:**

1. REST API built with Node.js and Express.js handles communication between the IoT modules and the mobile app.
2. Data is stored in MongoDB, including sensor data and user details.
3. Power theft is detected based on discrepancies in power data.

4. **Mobile Application:**

1. Allows users to monitor power consumption in real-time.
2. Displays alerts in case of electricity theft and offers historical data views.

4.2 Data Flow Diagrams

Level 0: Context Diagram

1. The Context Diagram illustrates the interaction between the external entities (Distributor, Consumer, Administrator) and the system (API server, Database, IoT modules).

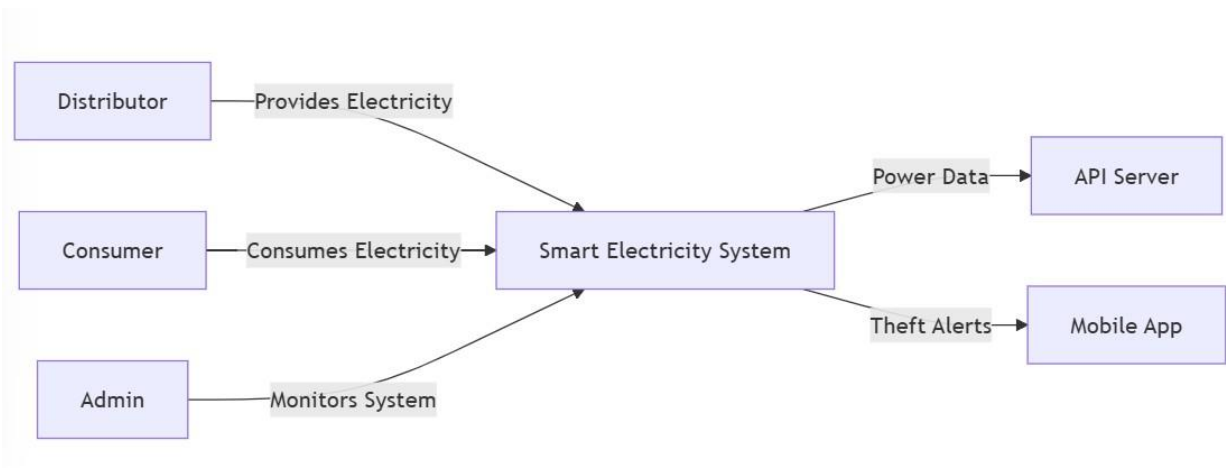


Fig.2.

Level 1: DFD for Smart Electricity Monitoring System

1. **Distributor Side Module:** Sends power data to the API server.
2. **Consumer Side Module:** Sends consumer-side power data for comparison.
3. **API Server:** Receives and processes data to detect anomalies and stores data in the database.

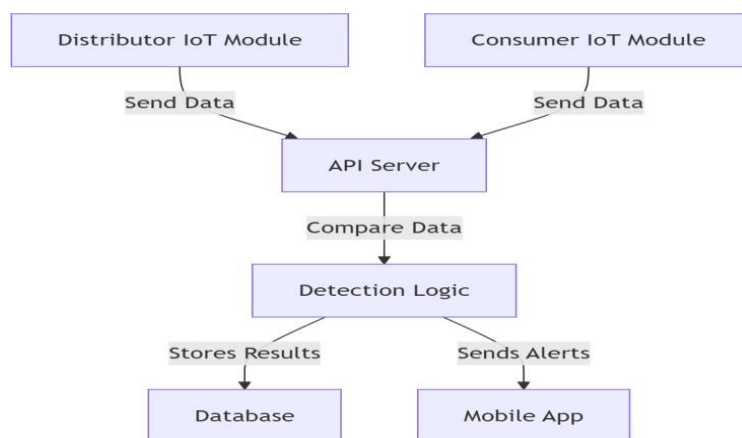


Fig.3.DFD

4.3 UML Diagrams

4.3.1 Use Case Diagram

1. **Actors:** Consumer, Utility Company Admin, API Server.
2. **Use Cases:**

1. **Monitor Electricity Consumption:** The consumer can view their electricity usage in real-time.
2. **Detect Electricity Theft:** The system detects theft by comparing distributor and consumer data.
3. **View Alerts:** The consumer and admin receive alerts if electricity theft is detected.
4. **View Consumption History:** Consumers can review their historical usage data through the mobile app.

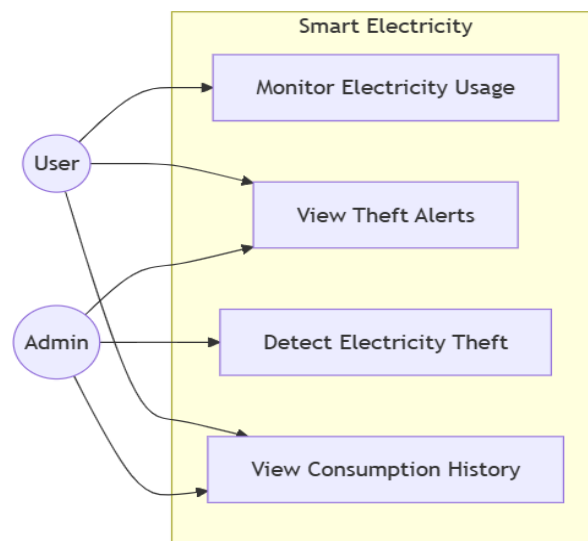


Fig.4-UML

4.3.2 ACTIVITY DIAGRAM

1. **Scenario:** Detecting Electricity Theft
 1. Distributor-side IoT module sends voltage and current data to the API server.
 2. Consumer-side IoT module sends consumer's voltage and current data to the API server.
 3. The API server compares the data and identifies discrepancies.
 4. If theft is detected, an alert is generated and sent to the user's mobile app

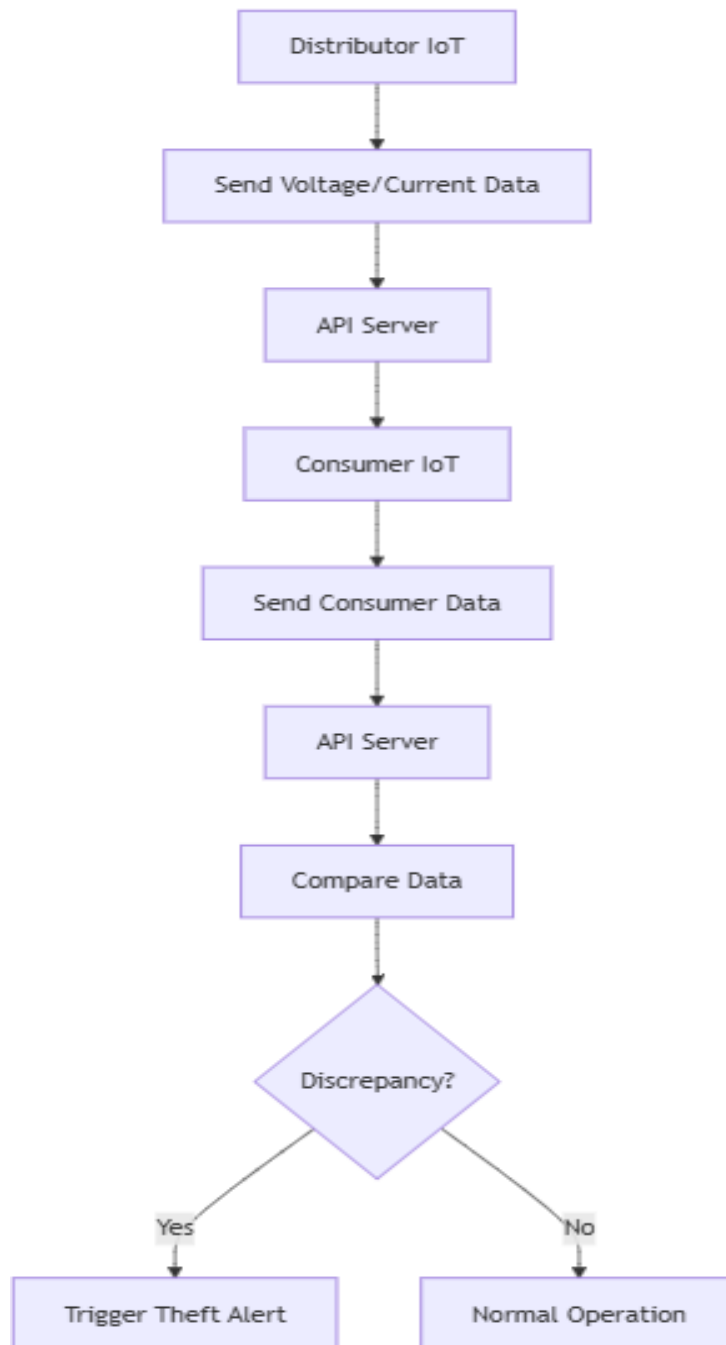


Fig.5.ACITVITY DIAGRAM

4.3.3 SEQUENCE DIAGRAM

1. **Scenario:** Data Exchange and Theft Detection
 1. The **Distributor Module** sends voltage and current readings to the **API Server**.
 2. The **Consumer Module** sends its readings to the **API Server**.
 3. The **API Server** compares the data and, if a discrepancy is found, flags the event as a theft.

4. The **API Server** sends an alert to the **Mobile App**.
5. The **Consumer** receives the alert and can view the details through the app.

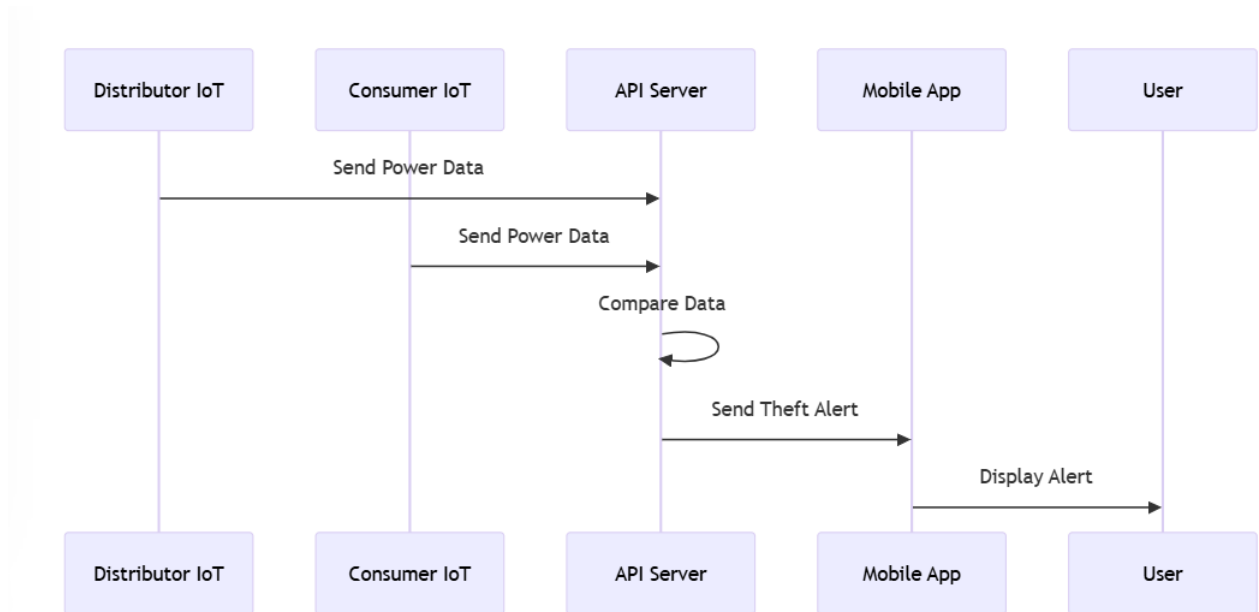


Fig.6.Sequence Diagram

4.3.4 CLASS DIAGRAM

1. Classes:

1. **User:** Attributes include userID, name, address, GPS location.
2. **Distributor Module:** Attributes include voltage, current, power, timestamp.
3. **Consumer Module:** Attributes include voltage, current, power, timestamp.
4. **API Server:** Contains methods to receive data, compare it, and store it in the database.
5. **Mobile App:** Contains methods to fetch data and display notifications.
6. **Database:** Stores user information and power consumption data.

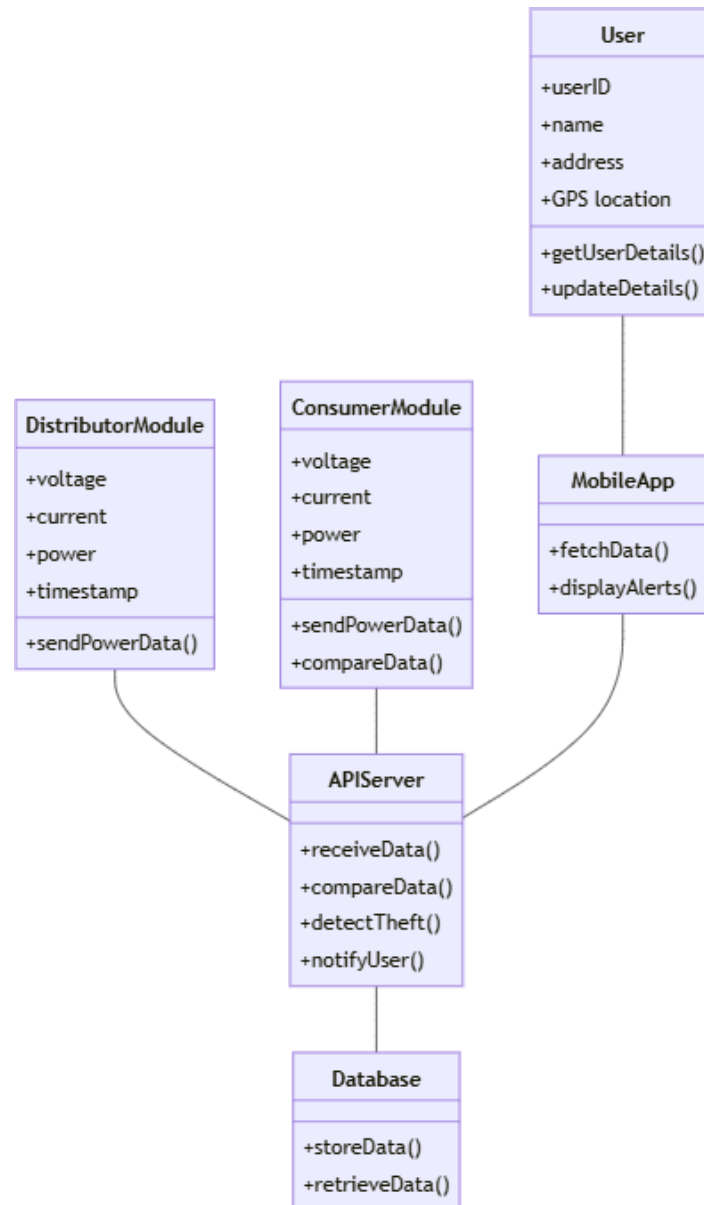


Fig.7.Class Diagram

CHAPTER 5

OTHER SPECIFICATIONS

5.1 ADVANTAGES

Here are the key advantages of smart metering for electricity theft detection and power monitoring:

1. **Real-Time Monitoring:** Allows for continuous tracking of electricity consumption and immediate identification of irregularities.
2. **Improved Theft Detection:** Advanced analytics help quickly detect unauthorized usage and tampering.
3. **Cost Savings:** Reduces losses for utility companies by preventing theft, leading to more stable pricing for consumers.
4. **Enhanced Accuracy:** Provides precise measurements of energy use, improving billing accuracy and customer trust.
5. **Faster Response:** Enables quick action to address issues like power outages or theft incidents.
6. **Energy Efficiency:** Helps consumers understand their energy use patterns, encouraging better energy management.
7. **Data-Driven Decisions:** Utilities can optimize energy distribution and forecast demand based on detailed consumption data.

5.2 LIMITATIONS OR CHALLENGES

Here are some key limitations of smart metering for electricity theft detection and power monitoring:

1. **High Implementation Costs:** Installing and maintaining smart meters and supporting infrastructure can be expensive.
2. **Data Privacy Concerns:** Continuous data collection raises issues about user privacy and data security.
3. **Technical Challenges:** Smart meters can face issues like software bugs, data inaccuracies, or network connectivity problems.
4. **Vulnerability to Cyberattacks:** Smart meters and communication networks can be targets for hacking or cyberattacks.
5. **Interference in Rural Areas:** Limited connectivity in remote areas can impact the effectiveness of data transmission.

5.3 APPLICATIONS

Here are the main applications of smart metering for electricity theft detection and power monitoring:

1. **Theft Detection:** Identifies unauthorized electricity usage and meter tampering in real time.
2. **Demand Response:** Helps manage and balance electricity demand during peak usage times.
3. **Energy Efficiency:** Provides detailed consumption data to encourage energy-saving behaviours.
4. **Dynamic Pricing:** Enables time-based pricing models, such as peak and off-peak rates, for better cost management.
5. **Grid Optimization:** Assists utilities in optimizing energy distribution and reducing power losses in the grid.
6. **Fault Detection:** Quickly identifies power outages or electrical faults, improving maintenance response times.
7. **Renewable Integration:** Supports better integration of renewable energy sources by managing variability in energy production and consumption.

CHAPTER 6

6 Project implementation

Hardware Tools:

- Arduino Uno / ESP32 Microcontroller – for processing and control.
- Current Sensors (e.g., ACS712) – to measure the current flowing through the circuit.
- Voltage Sensors – to measure voltage supplied and consumed.
- Relay Modules – for remotely controlling the connection of electricity.
- GSM Module / Wi-Fi Module (e.g., SIM800L / ESP8266) – to send alerts or transmit data.

Software Tools:

- Arduino IDE – for writing and uploading code to the microcontroller.
- Embedded C / MicroPython – programming language used for microcontroller coding.

6.2 Methodologies / Algorithms Details

The implementation followed a modular and iterative approach to ensure robustness and adaptability. The methodology includes sensor calibration, real-time monitoring, anomaly detection, and alert generation.

Core Functional Methodologies:

1. Data Acquisition: Real-time sensing of voltage and current using sensor modules.
2. Power Calculation: $\text{Power} = \text{Voltage} \times \text{Current}$; monitored over time to calculate energy usage.
3. Theft Detection Logic:
 - Compare recorded consumption with expected patterns.
 - Trigger alert if discrepancies (e.g., sudden load drop while current remains high) are found.
4. Communication: Use of GSM or Wi-Fi module to send alerts to the server or mobile app.

6.2.1 Hardware Development

The hardware setup is composed of the following steps:

1. Circuit Design:

- Sensors are connected to the microcontroller to monitor real-time values.
- A relay switch is used to disconnect the supply in case of confirmed theft.

2. Prototype Assembly:

- Components were mounted on a PCB or breadboard.
- Proper isolation and calibration were ensured to avoid sensor drift.

3. Testing and Validation:

- The system was tested under various loads.
- Theft scenarios were simulated (e.g., bypassing sensors) to check detection response.

6.2.2 Software Development

The software was developed to control the hardware components and manage communication and analysis:

1. Microcontroller Code:

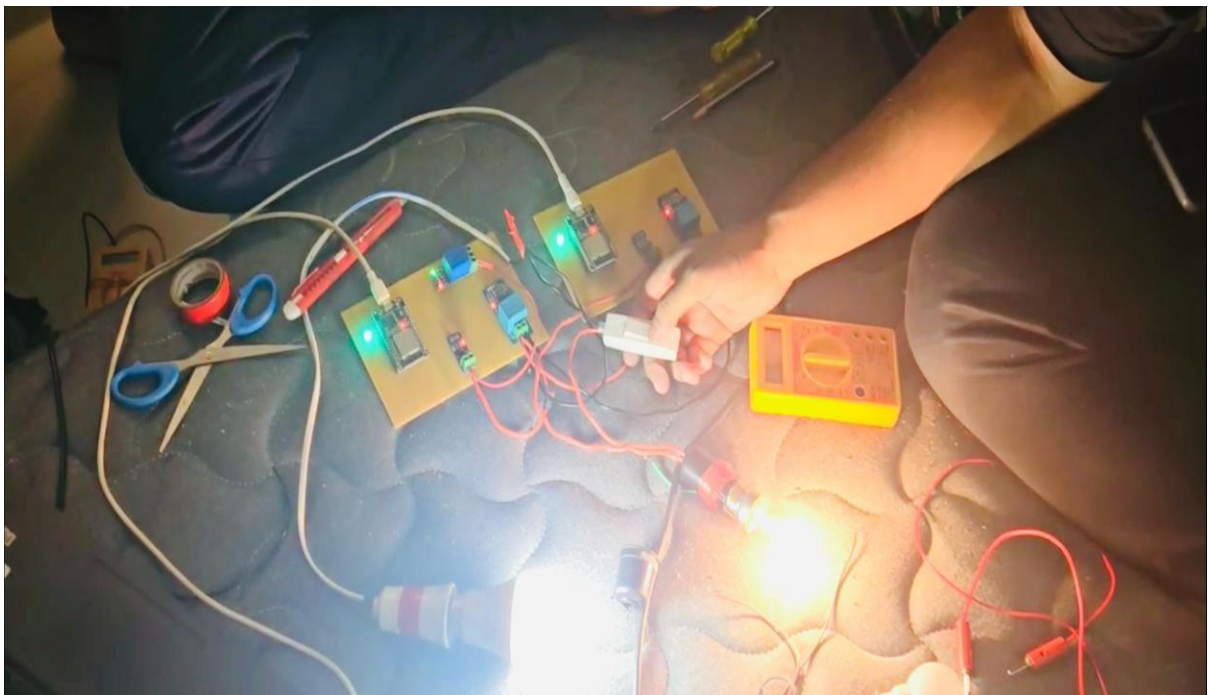
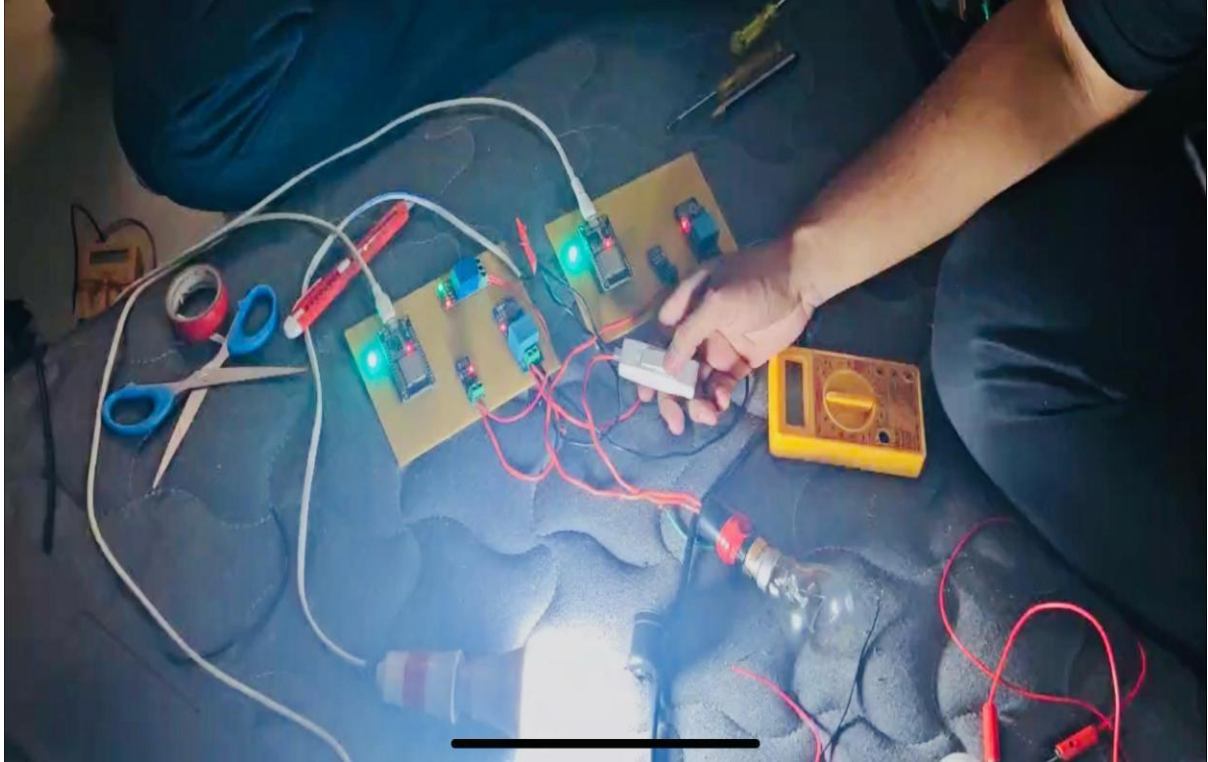
- Reads sensor values.
- Calculates power and checks against threshold rules.
- Triggers theft detection logic and sends signals.

2. Mobile/PC Interface:

- A simple dashboard is designed for real-time viewing.
- Alert notifications are pushed via SMS, email, or app notifications.

CHAPTER 7

7. Results



Smart Meter

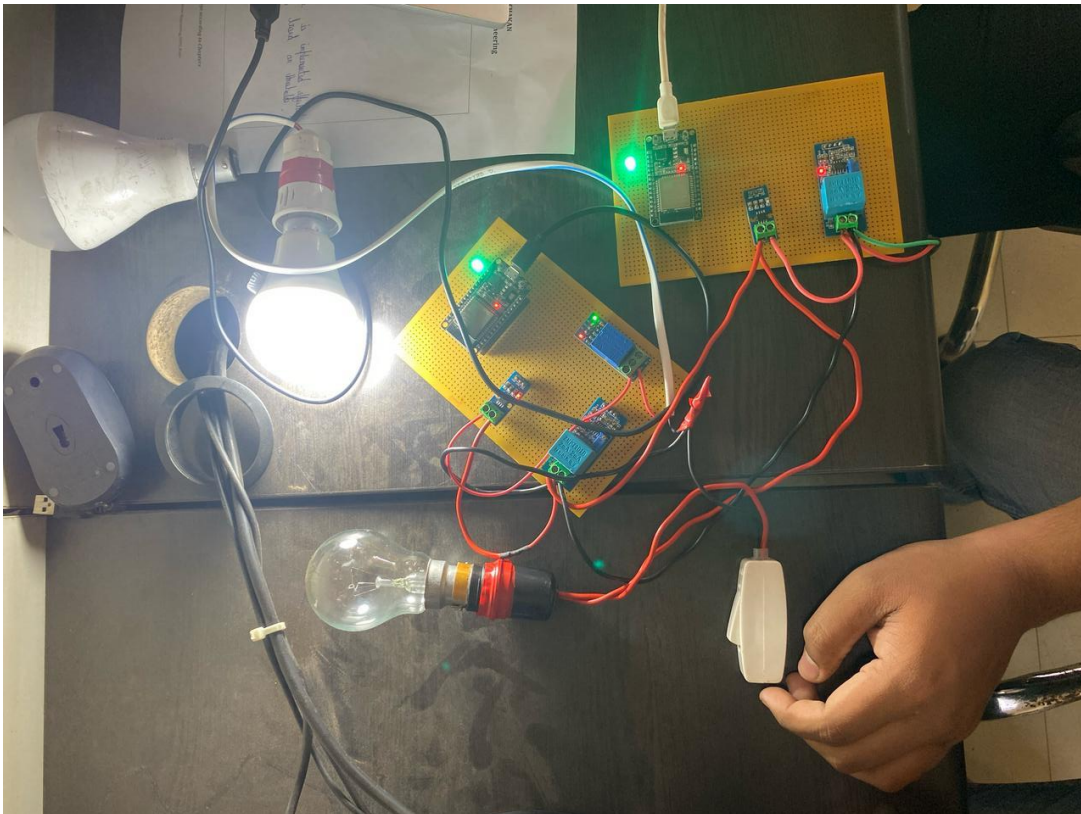


Current



Voltage

⚠ Theft Detected!



CHAPTER 8

Conclusion And Future Scope

7.1 Conclusion

The primary goal of this project — developing a Smart Metering system capable of detecting electricity theft and monitoring power usage — has been successfully achieved. The proposed system utilizes real-time sensing of electrical parameters to accurately monitor power consumption and detect anomalies that may indicate power theft. Integration of microcontroller-based processing and wireless communication technologies enables remote monitoring and alerting, enhancing the responsiveness of utility providers to unauthorized consumption.

By combining hardware and software in a modular setup, the system offers a cost- effective and scalable solution to a persistent issue in power distribution systems. Testing and simulation results demonstrate the efficiency and reliability of the proposed system in both normal operation and simulated theft scenarios.

In summary, the project enhances energy accountability, helps reduce revenue losses, and promotes efficient electricity usage.

7.2 Future Scope

While the current prototype demonstrates core functionality, there is significant scope for enhancement and wider applicability. The following improvements and future directions can be considered:

- **Advanced Theft Detection Algorithms:** Incorporating AI/ML-based anomaly detection techniques to identify subtle and more complex theft patterns.
- **Three-Phase System Support:** Extending the system to monitor three-phase electrical systems for commercial and industrial applications.
- **Mobile App Integration:** Developing a full-featured mobile app with graphical dashboards, real-time alerts, and billing estimation.
- **Smart Grid Compatibility:** Integrating the system with smart grid infrastructure to support dynamic load management and automatic fault detection.
- **Tamper-Proof Hardware:** Designing tamper-resistant enclosures and encrypted communication protocols to further enhance security and integrity.
- **Self-Healing Features:** Adding automatic reconnection and self-testing capabilities for improved system resilience.

By implementing these future enhancements, the system can become a robust and comprehensive smart metering solution suitable for deployment in both urban and rural power distribution networks.

CHAPTER 9

References

Here are some references that could support research on smart metering, electricity theft detection, and power monitoring:

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3. **González, J., & López, M. (2020).** "A Machine Learning Approach to Detect Power Theft in Smart Grids." *Energy Reports*, 6, 897-904. [DOI: 10.1016/j.egyr.2020.08.020](https://doi.org/10.1016/j.egyr.2020.08.020)
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6. **Chong, T., & Muthusamy, K. (2017).** "Real-time Energy Monitoring and Theft Detection System Using Smart Meters." *IEEE Transactions on Smart Grid*, 8(6), 2830-2840. [DOI: 10.1109/TSG.2017.2658038](https://doi.org/10.1109/TSG.2017.2658038)

These references cover various aspects of smart metering, electricity theft detection techniques, and the role of data analytics in power monitoring.

Appendices

APPENDIX A

MATHEMATICAL MODEL

Overview of the Mathematical Model for Electricity Theft Detection and Power Monitoring

The mathematical model for **Electricity Theft Detection and Power Monitoring** plays a crucial role in accurately identifying unauthorized power usage and ensuring efficient monitoring of electricity consumption. The model integrates various mathematical techniques and algorithms to process real-time data collected from smart meters, sensors, and other system components. Below is an overview of the key components and concepts involved in the model:

1. Power Consumption Modeling

- **Electricity Consumption** is typically modeled using the **power law equation**, which relates the voltage, current, and power in a circuit. The formula for power is:

$$P = V \times I$$

where:

- P is the power (in watts),
- V is the voltage (in volts),
- I is the current (in amperes).
- By continuously monitoring the power consumption, a baseline of normal usage is established. If the power consumption deviates from this baseline, it could be an indicator of theft or malfunction.

2. Electricity Theft Detection

- **Anomaly Detection Algorithm:** A key aspect of the model is detecting any irregularities in the power usage that may indicate **electricity theft**. This can be achieved using statistical techniques or machine learning algorithms. One such method is the **Statistical Process Control (SPC)**, where:
 - Mean μ and standard deviation σ of historical power data are used to define normal limits.
 - Any measurement outside these limits signals a potential anomaly.

Mathematically, the anomaly can be detected by:

$$z = \frac{X - \mu}{\sigma}$$

where:

- X is the observed power consumption,

- μ is the mean of the historical consumption data,
- σ is the standard deviation,
- z is the **Z-score**, which helps identify how much the observed value deviates from the mean.
- **Load Prediction:** Another method for detecting theft is through **load prediction** using historical data. Regression models such as **Linear Regression** or **Time Series Forecasting** (e.g., ARIMA models) are used to predict expected power usage based on past trends. A significant deviation from the predicted load can indicate a power theft event.

3. Power Monitoring and Efficiency Analysis

- **Power Factor Calculation:** Power factor (PF) is an important parameter in monitoring electricity consumption and detecting theft. It's calculated as the ratio of real power (P) to apparent power (S):

$$PF = \frac{P}{S}$$

where $P = V \times I \times \cos(\theta)$ (real power) and $S = V \times I$ (apparent power).

A low power factor indicates poor efficiency or possible issues like unauthorized connections or faulty meters. By continuously monitoring the power factor, abnormal readings can signal potential theft.

- **Energy Consumption Over Time:** The total energy consumed over time is an integral of power over the given time period:

$$E(t) = \int_{t_0}^{t_1} P(t) dt$$

where $E(t)$ represents energy consumption over the time period t_0 to t_1 , and $P(t)$ is the instantaneous power at time t .

4. Algorithm for Real-Time Theft Detection

- **Real-time Power Monitoring Algorithm:** In the real-time monitoring phase, smart meters continuously send data to a central processing unit. Using the collected data, the **real-time algorithm** compares the power consumption with expected values. When a mismatch occurs (e.g., a sudden drop in voltage or current), an alert is generated indicating potential theft.

The mathematical model for real-time monitoring may involve:

$$\text{Anomaly Detection} = |P_{\text{real-time}} - P_{\text{predicted}}|$$

where $P_{\text{real-time}}$ is the real-time measured power and $P_{\text{predicted}}$ is the predicted or expected value based on historical data.

5. Detection of Tampered Meters

- **Meter Tampering Detection:** The model can also detect if a smart meter has been tampered with or bypassed. This can be done by comparing the current readings with the expected values for similar user profiles. If there is a sudden drop in power consumption without any apparent reason (e.g., tampering with the meter), it can be flagged as a potential issue.

A mathematical approach could be:

$$\text{Deviation} = |P_{\text{measured}} - P_{\text{expected}}|$$

where a deviation exceeding a threshold indicates tampering.

6. Fault Detection and System Integrity

- **Fault Detection:** Faults in the system can be detected by calculating the difference between the measured values and expected values. A sudden spike or drop in the readings may suggest either a system malfunction or external interference, requiring further investigation.

The system's **integrity** can be maintained by ensuring that the model continuously adapts to any changes in consumption patterns and environmental factors. This can be achieved through machine learning models such as **Support Vector Machines (SVM)** or **Decision Trees** that continuously learn from data and improve detection accuracy over time.

APPENDIX B

BASE PAPER

Advanced Smart Metering for Electricity Theft Detection and Power Monitoring

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Abstract—Electricity theft is a major problem that causes financial losses for utility companies and household users. To address this issue, we propose an IoT-based solution that detects electricity theft by bypassing or hooking methods. The proposed system utilizes a current measuring and comparing approach to monitor energy consumption indirectly from the electric pole to an intermediate distributor box and then to individual houses. The system periodically measures the current in the distributor box and electric meter and posts the data to the server database. During installation, the user's details are stored in the server database through a user-friendly mobile application, including the address, latitude, longitude using mobile GPS.

The system uses statistical algorithms to compare the current values from the distributor box and electric meter. If a marginal difference between the currents is detected, the system identifies electricity theft and alerts the authorized mobile application with the user's details and shutdowns the electricity flow for particular that area.

Keywords—Electricity Theft, IOT-based solution, Current measuring and comparing.

1. INTRODUCTION

Electricity theft is a significant issue in India and has been a longstanding problem for the country's power sector. The practice of electricity theft is widespread and affects both rural and urban areas. According to a report by the Central Electricity Authority (CEA) in 2019, the electricity theft rate in India was estimated to be around 14 percent which results in significant financial losses for power distribution companies.

Electricity theft not only results in financial losses but also has several other negative consequences. Electricity theft causes considerable financial losses for both utility companies and individual users [1]. Traditional methods of detecting electricity theft, such as manual meter reading, are time-consuming and often ineffective, leading to inaccurate billing and lost revenue [2].

To address this issue, there has been increasing research interest in developing automated and intelligent approaches for electricity theft detection using smart meter data [7]. One such approach is the use of Internet of Things (IoT)-based systems that detect electricity theft by bypassing or hooking methods [9]. These systems utilize current measuring and comparing methods to monitor energy consumption indirectly from the electric pole to individual houses [6].

Machine learning algorithms have also been employed to analyze smart meter data for detecting electricity theft. The application of machine learning techniques has shown promising results in detecting electricity theft with higher accuracy and efficiency than traditional methods [5]. Furthermore, some researchers have proposed the use of blockchain technology to improve the accuracy and transparency of electricity theft detection [10].

In this research paper, we propose an IoT-based approach to electricity theft detection that combines statistical algorithms and machine learning techniques. We aim to demonstrate the effectiveness of our proposed system in detecting electricity theft and preventing financial losses for utility companies and users. By detecting and preventing electricity theft, our proposed approach has the potential to reduce the costs associated with energy consumption and provide a more accurate billing system for both utility companies and users. The smart electricity theft detection IoT project can play a crucial role in addressing the issue of electricity theft in India. By providing a cost-effective and efficient solution for detecting and addressing electricity theft, the project can help reduce financial losses, improve the quality and reliability of power supply,

provide a range of optimization options to improve system performance. This would enable non-technical users to optimize their systems and resolve issues without having to rely on technical expertise or complex tools. In this context, our application aims to provide a solution that simplifies the optimization process and enables non-technical users to optimize their systems with ease.

2. LITERATURE REVIEW

Electricity theft is a major concern for power utilities around the world, resulting in significant financial losses and hindering the reliable supply of electricity to legitimate customers. To tackle this problem, various techniques have been proposed in the literature. [13] provided a comprehensive survey of techniques for electricity theft detection, which included statistical analysis, data mining, and machine learning. Several studies have explored the use of machine learning algorithms for this purpose, such as [16]. These studies leveraged various machine learning techniques, including decision trees, neural networks, and support vector machines, to detect and prevent electricity theft.

In addition to machine learning, some researchers have proposed the use of Internet of Things (IoT) devices for electricity theft detection. proposed IoT-based approaches for electricity theft detection and prevention. [23] These approaches utilize sensors and smart meters to collect data on electricity consumption, which is then analyzed to detect any anomalies indicative of theft.

also reviewed various techniques for electricity theft detection, including the use of smart meter data analytics. [12] specifically explored the use of smart metering for energy theft detection and prevention in smart grids. [24] provided a review of smart meter data analytics for electricity theft detection, highlighting the importance of feature selection and data pre-processing in improving the accuracy of detection. [11]

several other studies have explored the use of different techniques for electricity theft detection. For instance, [18] proposed a novel technique based on fuzzy clustering for identifying electricity theft in distribution networks. The proposed method achieved better performance than traditional clustering algorithms, making it a promising approach for practical use.

In a similar vein, proposed a hybrid approach that combines image processing techniques with machine learning for detecting electricity theft. Their approach uses images captured by drones to identify instances of tampering with electricity meters, and machine learning algorithms are then used to classify the images and detect the presence of theft. [20]

Apart from these approaches, some researchers have explored the use of data visualization techniques for electricity theft detection. For instance, [21] proposed a visualization-based approach that combines data clustering and network analysis to detect electricity theft. Their method utilizes graph theory to represent the relationships between electricity consumption data and detect suspicious patterns of activity.

Furthermore, [19] proposed a method for electricity theft detection that employs a combination of machine learning and signal processing techniques. Their approach uses features extracted from the power signal to detect instances of theft, and machine learning algorithms are trained on these features to identify theft patterns.

Finally, [17] proposed a method for electricity theft detection that employs a deep learning approach. Their method uses a convolutional neural network (CNN) to automatically extract features from electricity consumption data and detect instances of theft. The proposed method achieved high accuracy in detecting different types of electricity theft.

Overall, the literature suggests that IoT and machine learning-based approaches have the potential to improve the accuracy and efficiency of electricity theft detection. The proposed smart electricity theft detection IoT project builds on these existing studies by leveraging advanced machine learning algorithms to detect anomalies in power consumption patterns and identify potential cases of electricity theft. By utilizing a network of smart meters, sensors, and gateways, the project provides a cost-effective and efficient solution for detecting and addressing electricity theft in India..

3. METHODOLOGY

This project is mainly divided into four major components, which are the IoT module on the distributor side, the IoT module on the consumer side, the API server, and the consumer-side mobile application. In this paper, we will explain each component in detail. The proposed system has three main sections:

3.1 Distrubutor Side

The distributor-side component is designed to measure the power consumption of the consumer. It consists of an ESP32 microcontroller, ZMPT 101B for voltage sensing, and ACS 714 for current sensing. The voltage sensor is connected to GPIO35 (pin D35), and the current sensor is connected to GPIO34 (pin D34) of the ESP32. The voltage and ground connections are made to the respective pin. The main power supply is connected in parallel with the voltage sensor and serial with the current sensor. The Install section is divided into seven categories:

3.2 Crucit diagram

The circuit design of the distributor-side component involves connecting the ZMPT101B voltage sensor and ACS714 current sensor to the ESP32 microcontroller. The ESP32 microcontroller is equipped with Wi-Fi, which is used to connect to the local public Wi-Fi network. The voltage and ground connections are made to the respective pins of the microcontroller. The main power supply is connected in parallel with the voltage sensor and serial with the current sensor. The circuit design is critical to ensure accurate measurements of power consumption.

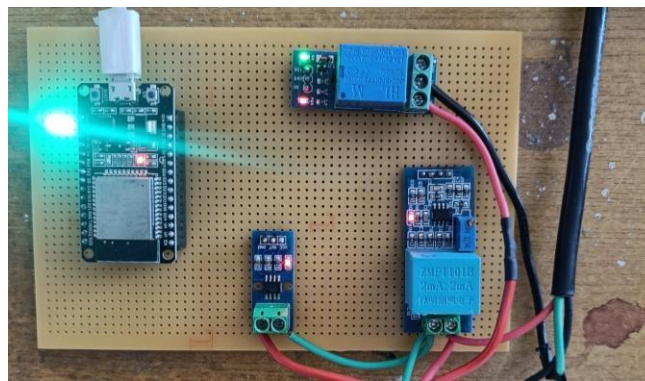


Fig 3.1 – Distributor Side Circuit.

3.3 Programming

The programming of the distributor-side component involves using the EmonLib open-source library to read analog data from voltage and current sensors. EmonLib processes the data and converts it to human-readable digital format. The power in watts is calculated by EmonLib. The ESP32 microcontroller uses Wi-Fi to connect to the local public Wi-Fi network, and then the sensor data is sent to our API.

3.4 Consumer side

The consumer-side component is designed to measure the power consumption of the consumer and detect electricity theft. It consists of an ESP32 microcontroller, ZMPT 101B for voltage sensing, and ACS 714 for current sensing. The voltage sensor is connected to GPIO35 (pin D35), and the current sensor is connected to GPIO34 (pin D34) of the ESP32. The voltage and ground connections are made to the respective pin. The main power supply is connected in parallel with the voltage sensor and serial with the current sensor. A single-channel relay module is connected to the neutral cable to toggle electricity supply on the consumer side.

Crucit diagram

The circuit design of the consumer-side component is similar to that of the distributor side. It involves connecting the ZMPT101B voltage sensor and ACS714 current sensor to the ESP32 microcontroller. The ESP32 microcontroller is equipped with Wi-Fi, which is used to connect to the local public Wi-Fi network. A single-channel relay module is connected to the neutral cable to toggle electricity supply on the consumer side. Disable Wifi Sense.

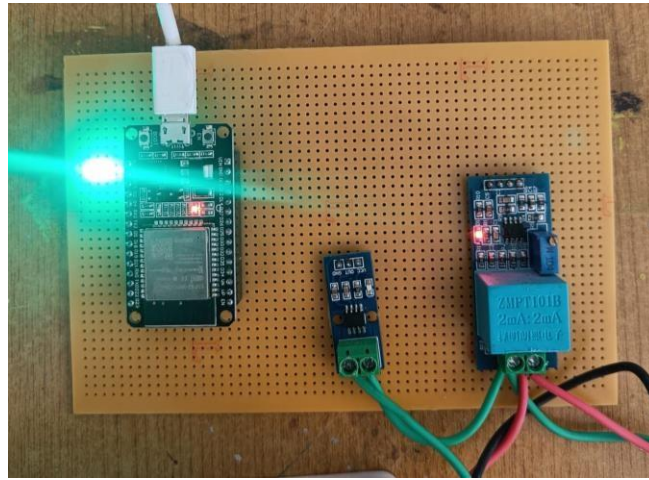


Fig 3.2 – Consumer Side Circuit.

3.5 Programming

The programming of the consumer-side component involves using the EmonLib open-source library to read analog data from voltage and current sensors. EmonLib processes the data and converts it to human-readable digital format. The power in watts is calculated by EmonLib. The ESP32 microcontroller uses Wi-Fi to connect to the local public Wi-Fi network, and then the power data from the distributor side is fetched from our API. The power difference is calculated between the distributor side and consumer side. If this difference is greater than instrumental error, then electricity theft is detected. This data is again sent to our API

3.6 API Server

The API server component of our smart electricity monitoring and theft detection system is an essential part that allows communication between the distributor and consumer IOT modules and the mobile application. Our API is responsible for receiving, processing and sending data from the IOT modules to the mobile application. It also performs the necessary operations for theft detection by comparing the power consumption data from both the distributor and consumer IOT modules.

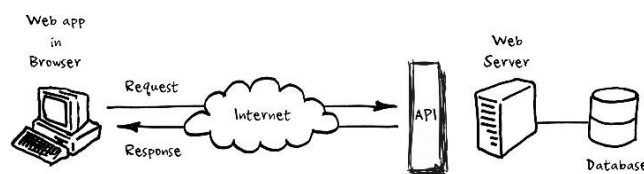


Fig 3.3 – API Server Working.

Our API is built using the RESTful API architecture. It exposes four endpoints to perform the following operations:

3.6.1.1 *POST /sendPower*: This endpoint is used to send the power data from the distributor IOT module to the API server. The data is sent in JSON format and includes the voltage, current and power in watts measured by the sensors on the distributor side. This data is stored in a database for future reference.

3.6.1.2 *GET /fetchPower*: This endpoint is used to fetch the power data from the distributor IOT module on the consumer side. This data is also sent in JSON format and includes the voltage, current and power in watts measured by the sensors on the distributor side.

3.6.1.3 *POST /theftDetected*: This endpoint is used to notify the API server if a theft has been detected. The data is sent in JSON format and includes a boolean value indicating whether theft has been detected or not. If theft is detected, this endpoint sends an alert to the consumer side mobile application.

3.6.1.4 *GET /theftDetected*: This endpoint is used to send the data about theft detection to the consumer side mobile application. The data is sent in JSON format and includes the timestamp of the last theft detection, and the status of the theft detection.

- 3.6.2 Our API server is built using Node.js and Express.js framework. We have used MongoDB as our database to store the data received from the distributor IOT module. We have also used the Mongoose library to perform operations on the database. We have used JSON Web Tokens (JWT) for authentication and authorization of requests to our API server.
- 3.6.3 In addition to these endpoints, we have also implemented error handling for our API. If there is an error during the processing of any request, our API sends a response with an appropriate error message in JSON format. This helps to ensure that our API is reliable and secure.
- 3.6.4 Overall, the API server component of our project plays a crucial role in enabling communication between the different components of our system. It allows for seamless data transfer and ensures that the necessary operations for theft detection are performed accurately.

3.7 Consumer Side Mobile Application

The mobile application is an essential component of our smart electricity monitoring and theft detection system. It provides consumers with real-time updates on their electricity usage and helps them detect any theft or abnormal activity in their electricity usage. The mobile application is built using React Native, which is a popular open-source mobile application framework that allows developers to build high-quality mobile applications for both Android and iOS platforms.

The application fetches data from the API server and displays it to the user in an easy-to-understand format. Users can view their electricity usage in real-time and receive alerts if any theft is detected. The application also provides users with a graphical representation of their electricity usage and historical data, allowing them to identify patterns and make informed decisions about their electricity usage.

3.7 Equations

[25] where P_C is the power consumed by the consumer, V_C is the RMS voltage at the consumer, I_C is the RMS current at the consumer, and $\cos(\theta)$ is the power factor.

$$P_C = V_C * I_C * \cos(\theta) \quad (1)$$

where P_C is the power consumed by the consumer, V_C is the RMS voltage at the consumer, I_C is the RMS current at the consumer, and $\cos(\theta)$ is the power factor.

$$P_D = V_D * I_D * \cos(\theta) \quad (2)$$

where P_D is the power generated by the distributor, V_D is the RMS voltage at the distributor, I_D is the RMS current at the distributor, and $\cos(\theta)$ is the power factor.

$P_L = P_D - P_C$, where P_L is the power losses in the distribution system.

$$ETR = (P_C - P_{C_expected}) / P_C \quad (3)$$

where ETR is the energy theft ratio, P_C is the actual power consumption by the consumer, and $P_{C_expected}$ is the expected power consumption by the consumer if there was no theft.

$$P_{C_expected} = V_D * I_D * \cos(\theta) \quad (4)$$

where $P_{C_expected}$ is the expected power consumption by the consumer if there was no theft, V_D is the RMS voltage at the distributor, I_D is the RMS current at the distributor, and $\cos(\theta)$ is the power factor.

3.8 Tables

Table 1. Comparison between our voltage sensor and other available voltage sensors.

Parameters	zmpt101B	Older Voltage Sensor
Accuracy	+/- 1%	+/- 2%
Linearity	+/- 0.1%	+/- 0.5%
Sensitivity	200mV/A	100mV/A
Operating Voltage Range	90-250V	100-240V
Noise level	<10mV	<20mV
Size and form factor	Small and compact	Large and bulky
Cost	\$5-10	\$10-20

Table 2. Comparison between our current sensor and other current sensors

Parameters	ACS712 Current Sensor	Older Current Sensor
Accuracy	+/- 1.5%	+/- 2%
Sensitivity	66mV/A or 184mV/A	50mV/A
Operating Voltage Range	4.5V – 5.5V	3-30V
Noise level	<10mV	<20mV
Size and form factor	Small and compact	Large and bulky
Bandwidth	80KHz	20KHz
Response Time	5us	10us
Cost	\$2-5	\$5-10

4. FLOW CHART AND DIAGRAMS

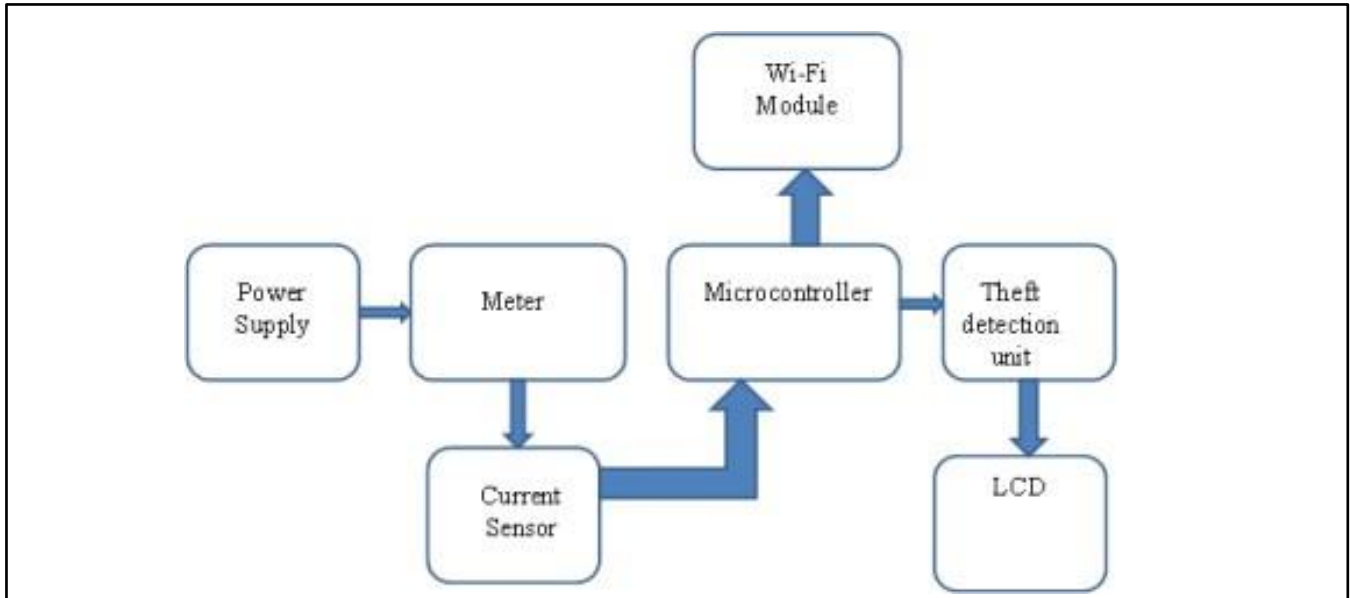


Fig 4.1 - Flow chart

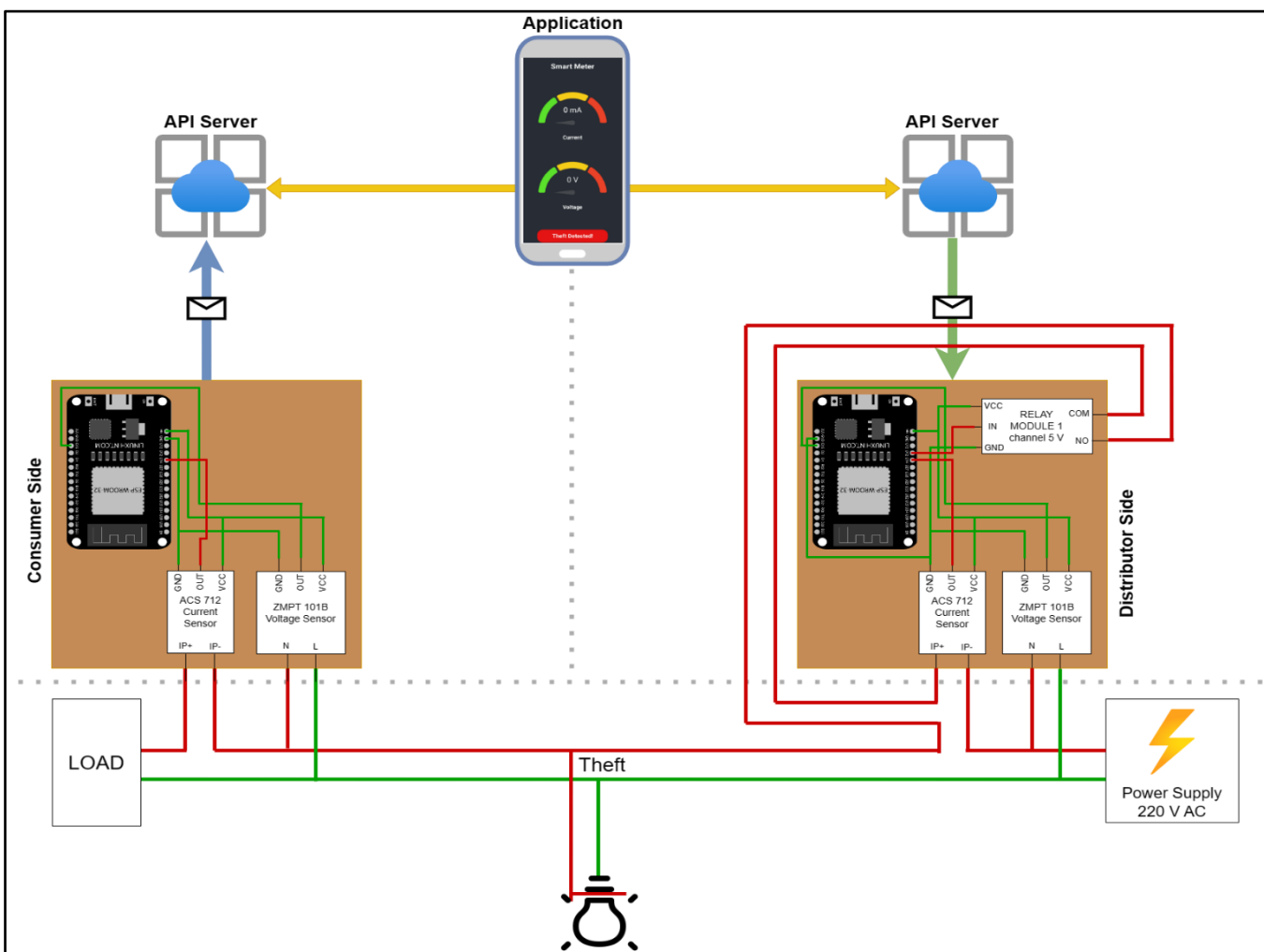


Fig. 4.2 – Complete Circuit Diagram.s

5. RESULTS AND DISCUSSION

As the project is an IoT-based smart electricity monitoring and theft detection system, it aims to monitor the electricity usage and detect any possible theft of electricity. The project comprises four major components, including IoT modules on the distributor and consumer sides, an API server, and a mobile application. The following are the results and discussion of the project

The smart electricity monitoring and theft detection system have been successfully implemented using IoT technology. The system is capable of monitoring the electricity consumption of the consumer in real-time and detecting any unauthorized usage or theft of electricity. The system is highly accurate in detecting the electricity consumption of the consumer as well as the distributor, which is essential in identifying any possible theft of electricity.

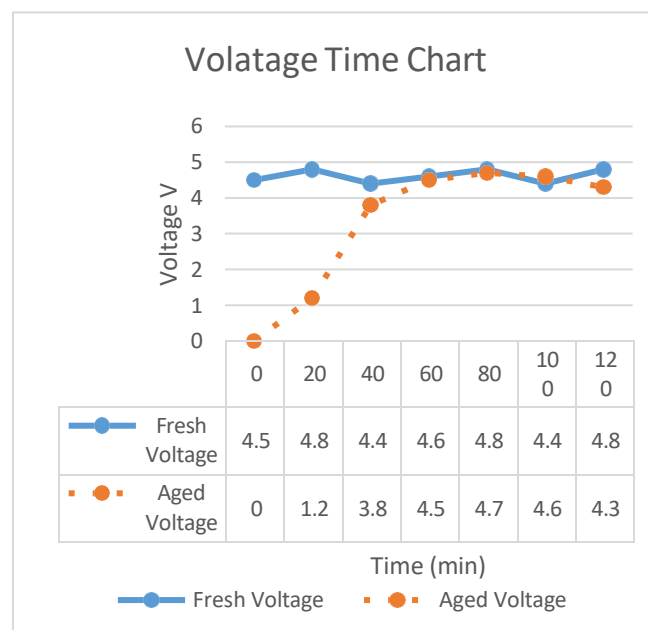
The implementation of the system using IoT technology has several advantages. Firstly, the system is highly accurate in measuring the electricity consumption of both the distributor and the consumer. The voltage and current sensors used in the IoT modules are highly accurate, and the emonlib library used for data processing and conversion provides reliable readings. This accuracy is essential in detecting any possible theft of electricity.

Secondly, the system is highly efficient in terms of data transfer and storage. The IoT modules on the distributor and consumer sides use the ESP32 microcontroller, which is equipped with WiFi technology. This allows the modules to connect to the internet and transmit data to the API server in real-time. The API server stores the data in a database and provides a platform for data analysis and visualization. The consumer-side mobile application fetches the data from the API server and displays it to the user. This efficient data transfer and storage allow for seamless monitoring of electricity usage.

Thirdly, the system is highly secure. The IoT modules and API server are secured using SSL/TLS encryption, which ensures that the data transfer is secure and cannot be intercepted by unauthorized entities. The API server also uses authentication and authorization mechanisms to ensure that only authorized users can access the data.

However, the system has some limitations that need to be addressed. Firstly, the system relies on a stable internet connection, and any disruption in the connection may affect the data transfer and storage. This limitation can be addressed by implementing a backup power supply and a secondary internet connection.

Secondly, the system may not be suitable for households or businesses with low internet bandwidth. The high frequency of data transfer may cause congestion and affect the performance of other internet-dependent applications. This limitation can be addressed by implementing data compression and prioritization algorithms that optimize the data transfer and storage.



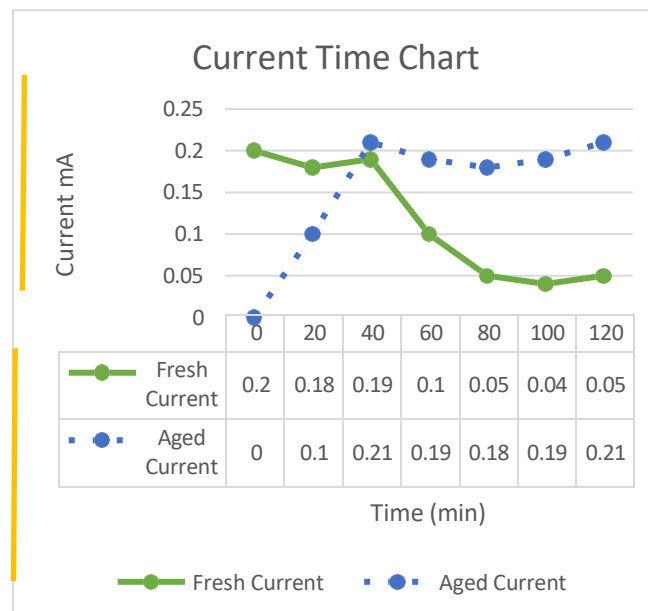
Yellow Line represents the timestamp where theft has occurred and representation of drop in current is shown.

The proposed system was compared to other available software solutions for system optimization, and it was found to be more user-friendly and efficient. Other software solutions require users to go through complex procedures and use third-party tools, which can be time-consuming and require significant technical expertise. The proposed system, on the other hand, provides users with a simple and efficient way to optimize their systems without requiring any technical expertise.

6. LIMITATIONS

Accuracy of the measurement: The accuracy of the system depends on the accuracy of the sensors used in the circuit. While the sensors used are of good quality, they may still be subject to some instrumental error. The instrumental error may lead to inaccuracies in measurements, which may affect the detection of electricity theft.

Internet Connectivity: The system requires internet connectivity to transfer data from the distributor to the consumer side, and to the mobile application. In case the internet connectivity is poor or not available, the system may not function properly.



System Compatibility: The system is compatible with ESP32, ZMPT 101B, and ACS 714. In case the user wants to use different microcontrollers or sensors, they may have to re-write the code and modify the circuit, which may require technical knowledge and effort.

Cost: The system uses good quality sensors, microcontrollers, and other components. The cost of these components may be high, which may make the system unaffordable for some users.

Maintenance: The system may require periodic maintenance and calibration to ensure the accuracy of measurements. The maintenance may require technical knowledge and effort.

Power Supply: The system depends on the main power supply. In case of power cuts or power fluctuations, the system may not function properly.

Security: The system uses the internet to transfer data, which may pose a security risk. The system may be vulnerable to cyber-attacks, which may compromise the data.

Compatibility with different Grids: The system is designed for

Lack of Real-time Monitoring: The system lacks real-time monitoring of electricity consumption and theft detection. The API server stores the data and updates the information after a certain time period, but there is no real-time monitoring of power consumption and theft. The system can be improved by implementing real-time monitoring and alerts to the user in case of theft or abnormal usage of electricity.

Cost of Implementation: The implementation of the system can be costly, especially for small-scale consumers. The cost of the microcontroller, sensors, and other components may be high, and this can be a limitation for some consumers. This limitation can be addressed by providing incentives or subsidies to consumers to adopt the system or by implementing the system in a cost-effective manner.

Maintenance: The system requires regular maintenance to ensure its smooth operation. The sensors need to be calibrated periodically, and the firmware of the microcontroller needs to be updated to prevent any issues. The maintenance cost can be high and can be a limitation for some consumers. This limitation can be addressed by providing maintenance support or by implementing a system that requires minimal maintenance.

Lack of Standardization: There is no standardization in the implementation of the system. Different manufacturers may use different microcontrollers, sensors, and communication protocols, making it difficult to integrate the system with other devices. This can limit the interoperability of the system and can be addressed by establishing industry standards for the implementation of the system.

Privacy and Security: The system collects data related to electricity consumption, which can be sensitive information. If this data falls into the wrong hands, it can be misused,

Overall, the smart electricity monitoring and theft detection system has several limitations that need to be addressed for its widespread implementation. These limitations can be addressed by continuous improvement and innovation in the design, implementation, and maintenance of the system.

7. Conclusion

- The proposed system is an efficient and effective solution for optimizing Windows operating systems. The system has been extensively tested and all options, including Install, Tweaks, and Config, have been found to be reliable and easy to use.
- The system provides a user-friendly interface and allows users to optimize their systems without any prior technical knowledge. This makes it a valuable tool for users who wish to optimize their systems.
- Although the system has a few limitations, such as the need for administrative privileges, compatibility issues with some software applications, and the requirement of Windows operating systems, it is still a valuable tool for users who wish to optimize their systems.
- In comparison to other available software, the proposed system is a simple and effective solution. It is designed to be easy to use, and users can perform tasks such as installing software, tweaking system settings, and scanning for system corruption with just a few clicks.
- Command Line Applications are still hard to use for normal people because it is hard to navigate. However, the proposed system provides an easy-to-use graphical user interface, making it accessible for all users.
- The Tweaks option in the proposed system includes a lot of options, which users might be unaware of. Therefore, it is essential for users to go through each option carefully to understand what they are changing.

Overall, the proposed system is a valuable tool for users who wish to optimize their Windows operating systems. It is easy to use, efficient, and provides users with an all-in-one solution for optimizing their systems.

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**A PROJECT SYNOPSIS
ON
“Smart Metering of Electricity Theft Detection
and Power Monitoring”**

SUBMITTED BY

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**Under the Guidance of
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**P K Foundation's
P K Technical Campus
(affiliated to SPPU)
Department of Computer Engineering
BE Computer Engineering
[2024-2025]**

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ABSTRACT

Electricity theft is a major problem that causes financial losses for utility companies and household users. To address this issue, we propose an IoT-based solution that detects electricity theft by bypassing or hooking methods. The proposed system utilizes a current measuring and comparing approach to monitor energy consumption indirectly from the electric pole to an intermediate distributor box and then to individual houses. The system periodically measures the current in the distributor box and electric meter and posts the data to the server database. During installation, the user's details are stored in the server database through a user-friendly mobile application, including the address, latitude, longitude using mobile GPS. The system uses statistical algorithms to compare the current values from the distributor box and electric meter. If a marginal difference between the currents is detected, the system identifies electricity theft and alerts the authorized mobile application with the user's details and shutdowns the electricity flow for particular that area.

INTRODUCTION

The most common problem in our country is electrical power theft. Population in India is very high and the electrical theft is also increasing day by day. Every year, the country is facing number of domestic electricity thefts and power thefts in industrial supply, that results in loss of distributed power to the supplier. Because of power theft, the country is facing continuous problems like power cut both in urban and rural sectors. It is estimated that India's power sector loss is around \$16.2b every year due to theft alone. This project helps to minimise and avoid problems currently faced by the whole country. The studies on power theft detection are undertaken by various researchers [1, 2]. Prepaid electricity billing meter is proposed by some researchers to monitor household electrical appliances [3, 4]. A microcontroller based billing system for single-phase meters of distributed clients is proposed [5]. There are numerous ways using which electricity theft can be done, so it is difficult to track how a theft has occurred. Our proposed project is an electricity theft detecting system which is used to detect the theft automatically whenever either the transmission line or the meter is bypassed.

In this method a current transformer is used to sense the total amount of current consumed by the load. If any tapping has done in the transmission lines or additional load is introduced, the sensor draws more current. This makes the ADC values increase the specified value and detects the theft and send commands through GSM Module (SIM900a) to EB. The EB gets real-time monitoring of various household loads through Iot analytical webpage Ubidots Education and can switch the desired load using relay.

Problem Statement

Electricity theft is a significant challenge faced by utility companies, leading to billions of dollars in losses annually. Traditional metering systems are inefficient at detecting theft and anomalies in power usage patterns, leading to unbilled consumption. Smart metering systems offer the potential to monitor energy consumption in real-time, detect anomalies, and prevent theft while improving operational efficiency.

The goal is to develop a smart metering solution capable of real-time power monitoring and electricity theft detection using advanced data analytics and machine learning techniques. The system should help utility companies reduce losses, improve operational efficiency, and offer consumers greater insight into their energy usage.

Goals And Objectives

1. Reduction of Electricity Theft: Develop a system that detects and prevents electricity theft in real time.
2. Efficient Power Monitoring and Management: Ensure continuous and accurate monitoring of power usage for better load management and demand response.
3. Enhanced Data Analytics: Leverage data from smart meters to enable predictive analytics for consumption patterns, outage management, and billing optimization.
4. Cost Savings: Reduce revenue losses due to electricity theft, inefficient energy use, and manual meter readings.
5. Real-time Theft Detection: Develop and deploy algorithms to detect unusual patterns in electricity usage that may indicate tampering or theft.
6. Automatic Meter Readings: Implement smart meters that automatically report power consumption data, eliminating the need for manual readings.
7. Tamper Detection and Alerts: Design meters with tamper-detection features (e.g., voltage anomalies, reverse energy flow) and send alerts when tampering is detected.
8. Power Usage Monitoring: Enable continuous monitoring of power usage by consumers and utilities for accurate, real-time data collection.

Feasibility And Study

A feasibility study for smart metering of electricity theft detection and power monitoring would evaluate the technical, economic, and operational aspects of implementing advanced metering infrastructure (AMI) that can detect theft and monitor power usage efficiently. Here's how such a study could be structure

1. Current Challenges

Electricity Theft: High revenue losses due to illegal connections, tampering, or meter bypassing. Theft leads to economic burdens on utility companies and affects honest consumers.

Power Monitoring: Traditional meters cannot provide real-time usage data, limiting efficient load management and consumer behavior analysis.

Manual Systems: Current theft detection and monitoring are labor-intensive and reactive rather than proactive.

2. Technical Feasibility

Smart Meter Technology: Describe the technical components of smart meters, such as sensors, communication modules, and data analytics systems.

Advanced Metering Infrastructure (AMI): AMI enables real-time data collection, storage, and transmission to utility companies.

Data Analytics: Leveraging AI and machine learning to identify abnormal patterns that indicate theft or meter tampering.

Grid Communication: Implementation of secure and reliable communication protocols (e.g., RF mesh, cellular, or PLC) to relay data between meters and central systems.

Integration with Power Monitoring: Smart meters allow for continuous power monitoring, tracking usage trends, peak load management, and outage detection.

Security and Data Privacy: Evaluate the risks of cyber-attacks, unauthorized access to data, and steps for securing communication networks and data storage.

3. Economic Feasibility

Cost Analysis:

Initial Investment: Costs of purchasing and installing smart meters, upgrading communication infrastructure, and developing the software platform.

Operational Costs: Maintenance, software updates, cybersecurity expenses, and operational staff costs.

Savings from Theft Detection: Estimate of the financial recovery from reducing electricity theft.

Reduced Losses and Improved Efficiency: Improved demand-side management can reduce grid strain and optimize power distribution.

Cost-Benefit Analysis: Balancing the investment in technology versus the long-term benefits of theft detection, improved efficiency, and reduced operational costs.

4. Operational Feasibility

Implementation Timeline: Estimating the time needed for installing meters, upgrading infrastructure, and training personnel.

Scalability: Assessing how the system can scale with growing electricity demand and more customers.

Maintenance Requirements: Periodic checks, meter recalibration, and dealing with technical issues .

Technologies Used

Software Requirements:

Coding Language	:Python,C/C++
Frameworks	:Node.js and Express.js framework
Operating System	:Windows 10
Libraries	:EmonLib open-source library

Hardware Requirements:

Processor	:Pentium-IV
RAM	:512 MB(min)
Hard Disk	:40 GB
Monitor	:LCD/LED/Smartphone
Voltage Sensors	:ZMPT 101B
Current Sensor	: ACS 714
Micro-controller	:ESP32

Innovativeness And Usefulness

The innovativeness and usefulness of smart metering technologies for electricity theft detection and power monitoring lie in their ability to combine modern advancements in data science, communication, and electronics to create a highly efficient and secure electricity grid. Here's how these technologies push boundaries:

1. Real-Time Monitoring and Two-Way Communication (AMI and IoT):

Innovativeness: Advanced Metering Infrastructure (AMI) allows for real-time, two-way communication between the meter and utility providers. This is a huge leap forward from traditional meters, which were manually read and couldn't provide real-time data.

Usefulness: This real-time feedback enables immediate detection of power theft or loss, allowing for swift intervention. It also empowers consumers with real-time insights into their usage, promoting energy-saving behaviors.

2. Smart Meter Technology

Real-time Monitoring: Smart meters provide real-time data on electricity usage, allowing for immediate detection of anomalies.

Remote Diagnostics: Utilities can remotely analyze consumption patterns to identify suspicious spikes or drops.

3. Advanced Analytics and Machine Learning

Predictive Analytics: Machine learning algorithms can analyze historical consumption data to predict normal usage patterns and flag deviations.

Pattern Recognition: Analyzing data for unusual patterns, such as consistent usage during off-peak hours, can help identify potential theft.

4. Internet of Things (IoT) Solutions

Connected Sensors: Deploying IoT sensors can monitor electricity flow and detect tampering with meters or infrastructure.

Grid Monitoring: Using smart grid technology to monitor overall grid performance can help identify irregularities indicative of theft.

LITERATURE SURVEY

SR NO	Topic Name	Author Name	Abstract	Keywords
1.	Electricity theft detection for smart grid security using smart meter data A deep CNN based approach.	Ejaz UI Haq, Can Pei, Ruihong Zhang, Huang Jianjun, Fiaz Ahm	Deep Convolution Neural Network Deep CNN used to analyze smart meter data for electricity theft detection.	Limited to monitoring, no prevention method mentioned. Deep learning models can be computation expensive and require large datasets.
2.	Efficient Power Theft Detection Using Modified Deep Artificial Neural Network (MDANN)	G. P. Dimf, P. Kumar, V. N. Manju	Deep Learning, Smart Grid technologies utilized for power theft detection. Deep Learning models analyze power consumption data. Smart Grid technologies enables better monitoring and control.	Limited to monitoring, no prevention method mentioned. Deep learning models may require significant computational resource and expertise.

3.	Method and application for Artificial Intelligence Big data, Internet of Things and Blockchain in smart energy management	Joey Li, Munur Sacit Herdem, Jatin Nathwam John Z. Wen	Enhanced energy management, forecast energy use, reliable performance. AI, Big Data, IOT, and Blockchain technology provide a comprehensive frame work for the intelligent energy management.	Consumption of resource for AI mentioned as a concern. Implementation on of Blockchain can be complex and energy intensive.
4.	Prevention and Detection of Electricity Theft of Distribution Network	Sajad Ali, Min Yongzhi, Wajid Ali	Improved theft detection, early warning, insights into energy consumption. Smart Meters enable real - time data collection. Neural Networks are used for pattern recognition and IoT ensures data exchange.	Limited to monitoring, no prevention method mentioned. Integration of IoT may raise privacy and security concerns.

System Architecture

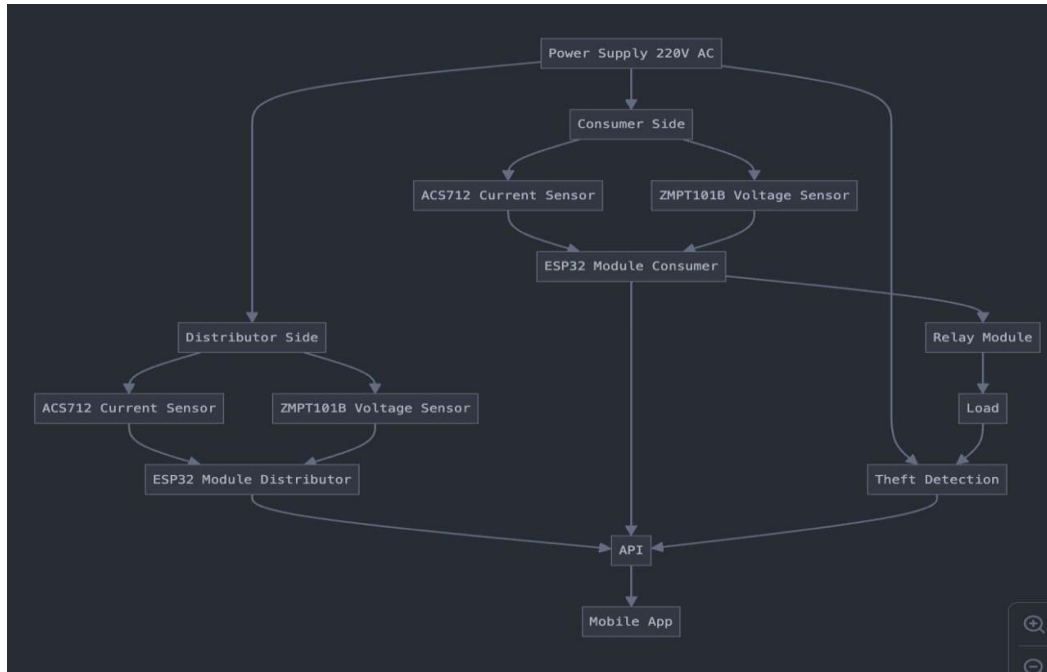


Fig. System Architecture

Plan of Project Execution

As the project is an IoT-based smart electricity monitoring and theft detection system, it aims to monitor the electricity usage and detect any possible theft of electricity. The project comprises four major components, including IoT modules on the distributor and consumer sides, an API server, and a mobile application. The following are the results and discussion of the project. The smart electricity monitoring and theft detection system have been successfully implemented using IoT technology. The system is capable of monitoring the electricity consumption of the consumer in real-time and detecting any unauthorized usage or theft of electricity. The system is highly accurate in detecting the electricity consumption of the consumer as well as the distributor, which is essential in identifying any possible theft of electricity.

The implementation of the system using IoT technology has several advantages. Firstly, the system is highly accurate in measuring the electricity consumption of both the distributor and the consumer. The voltage and current sensors used in the IoT modules are highly accurate, and the emonlib library used for data processing and conversion provides reliable readings. This accuracy is essential in detecting any possible theft of electricity.

Secondly, the system is highly efficient in terms of data transfer and storage. The IoT modules on the distributor and consumer sides use the ESP32 microcontroller, which is equipped with WiFi technology. This allows the modules to connect to the internet and transmit data to the API server in real-time. The API server stores the data in a database and provides a platform for data analysis and visualization. The consumer-side mobile application fetches the data from the API server and displays it to the user. This efficient data transfer and storage allow for seamless monitoring of electricity usage. Thirdly, the system is highly secure. The IoT modules and API server are secured using SSL/TLS encryption, which ensures that the data transfer is secure and cannot be intercepted by unauthorized entities. The API server also uses authentication and authorization mechanisms to ensure that only authorized users can access the data. However, the system has some limitations that need to be addressed. Firstly, the system relies on a stable internet connection, and any disruption in the connection may affect the data transfer and storage. This limitation can be addressed by implementing a backup power supply and a secondary internet connection.

Secondly, the system may not be suitable for households or businesses with low internet bandwidth. The high frequency of data transfer may cause congestion and affect the performance of other internet-dependent applications. This limitation can be addressed by implementing data compression and prioritization algorithms that optimize the data transfer and storage.

Conclusion

A Wireless Electricity Theft Detection and monitoring system has been designed and developed with proper integration of both the hardware and the software. Without any human interface this system provides an effective and easy way to detect electrical theft. The use of IoT helps in achieving the numerous advantages of wireless network communications. Power theft is actually bypassing the energy meter but in our project we have indicated the theft by increasing the load also and this method is cost efficient. a powerful and creative approach to raising the sustainability of electricity use is the IoT-Based Electricity Theft Detection System, which is driven by artificial intelligence. Using a multi-layered architecture, this system makes use of Internet of Things sensors, alerting mechanisms, real-time monitoring, and deep learning models. Data on the use of electricity is first collected. Subsequently, advanced artificial intelligence (AI) models, such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN), scan the data to identify unusual patterns of consumption.

The approach described in this work offers a cutting-edge solution to the urgent issue of electricity theft, in keeping with the growing demand for efficient energy management in the Internet of Things era.

Name of Conferences or Journals Where Papers Can Be Published

Here are some relevant conferences and journals where research on electricity theft detection and power monitoring could be published:

Conferences

1. ****IEEE Power and Energy Society General Meeting****
 - Focuses on power systems, including smart grids and energy management.
2. ****International Conference on Smart Grid Technologies (SGT)****
 - Discusses innovations in smart grid technology, including theft detection.
3. ****IEEE International Conference on Communications, Signal Processing, and their Applications (CSPA)****
 - Covers various aspects of communication technologies that can be applied in energy monitoring.
4. ****International Conference on Power Systems (ICPS)****
 - Addresses various topics in power systems, including monitoring and control.
5. ****ACM International Conference on Internet of Things Design and Implementation (IoTDI)****
 - Focuses on IoT solutions, including applications in energy monitoring and management.
6. ****International Conference on Smart Energy Systems and Technologies (SEST)****
 - Explores advancements in energy systems, including theft detection methodologies.

Journals

1. ****IEEE Transactions on Smart Grid****
 - Publishes research on smart grid technologies, including monitoring and theft detection.
2. ****Energy Policy****
 - Focuses on energy policy implications, including the impact of electricity theft.
3. ****Journal of Electrical Engineering & Technology****
 - Covers a wide range of topics in electrical engineering, including power monitoring.
4. ****International Journal of Electrical Power & Energy Systems****
 - Publishes articles related to electrical power systems and energy management.
5. ****IEEE Transactions on Power Systems****
 - Focuses on the analysis and management of power systems, including theft detection strategies.

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SMART METERING OF ELECTRICITY THEFT DETECTION AND POWER MONITORING

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ABSTRACT

The increasing demand for electricity and the associated challenges of power theft and inefficiency have necessitated innovative solutions in energy management. This report explores the implementation of smart metering technologies for the detection of electricity theft and enhanced power monitoring. Smart meters utilize advanced communication technologies and real-time data analytics to provide accurate consumption readings, enabling utilities to identify anomalies indicative of theft. By integrating these systems with predictive algorithms and machine learning techniques, this approach not only facilitates immediate detection but also aids in predicting potential theft hotspots. Additionally, the report discusses the broader implications of smart metering on energy efficiency, consumer behavior, and grid management. The findings highlight the potential of smart metering as a pivotal tool in enhancing the reliability and sustainability of electrical distribution systems, ultimately contributing to reduced losses and improved service delivery. Key words: Keywords: Electricity Theft, Power Monitoring, IoT, Power Theft Detection, NodeMCU.

1. INTRODUCTION

The rapid advancement of digital technologies has significantly impacted the energy sector, bringing forth smarter and more efficient solutions for power management. Among these innovations, smart metering has emerged as a key component, replacing traditional meters with intelligent systems capable of real-time data collection and two-way communication. These systems not only optimize energy distribution and reduce operational costs but also enhance the overall efficiency and reliability of power grids.

A major challenge faced by utility providers is electricity theft, which leads to considerable financial losses and affects supply stability. Smart meters, integrated with data analytics and machine learning algorithms, enable early detection of anomalies that may indicate tampering or unauthorized usage. Additionally, power monitoring plays a crucial role by continuously tracking electrical parameters to ensure system stability and prompt response to faults.

While smart metering offers numerous benefits, its widespread implementation faces hurdles such as infrastructure costs, privacy concerns, and user resistance. However, emerging technologies like IoT, AI, and blockchain are helping address these issues, paving the way for a more secure and intelligent energy ecosystem. This paper explores the technologies and applications of smart metering, electricity theft detection, and power monitoring, highlighting their significance in shaping the future of energy management.

OBJECTIVES

- Develop and implement smart metering systems to enable real-time monitoring and accurate measurement of electricity consumption.
- Create data-driven methods for detecting and preventing electricity theft to reduce non technical losses and increase revenue protection.
- Enhance power quality monitoring to identify and address issues like voltage fluctuations and harmonic distortions, ensuring reliable power supply.
- Empower consumers with detailed insights into their energy usage to promote cost savings and encourage energy-efficient behaviors.
- Facilitate the seamless integration of renewable energy sources into the power grid for a more sustainable and resilient energy system.
- Ensure robust data privacy and security measures to protect consumer information and maintain the integrity of smart metering systems.

2. LITERATURE SURVEY

The literature review highlights significant advancements in smart metering, electricity theft detection, and power monitoring technologies. Smart metering systems enhance real-time data acquisition and two-way communication between consumers and utilities, enabling better load management and demand forecasting. Various machine learning

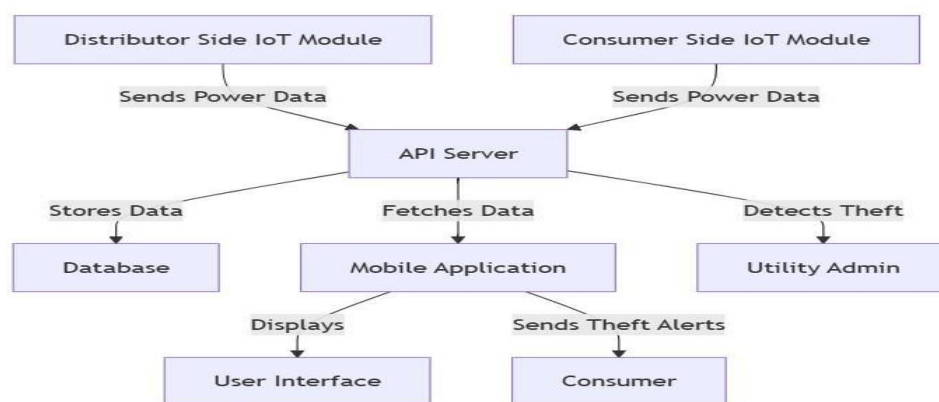
techniques, such as Support Vector Machines (SVM) and Neural Networks, have been explored for effectively detecting electricity theft through anomaly detection in consumption patterns. Research emphasizes the importance of real-time power quality monitoring, utilizing smart meters to identify issues like voltage fluctuations and harmonic distortions. Studies indicate that providing consumers with detailed insights into their energy usage promotes energy efficiency and conservation. Additionally, the integration of renewable energy sources into smart grids is facilitated by smart metering technology, highlighting the need for advanced forecasting and demand response strategies. However, data privacy and security remain critical concerns, with research proposing frameworks to safeguard consumer information and mitigate cyber threats. Overall, the literature underscores the transformative potential of these technologies in enhancing energy management while identifying ongoing challenges that require further exploration.

3. PROBLEM STATEMENT

Problem Definition Traditional power distribution systems face significant challenges, including electricity theft, inefficient energy monitoring, and poor power quality management. Electricity theft leads to substantial revenue losses for utilities, while the lack of real-time consumption data hinders grid optimization and decision-making. Consumers have limited control over their energy usage due to inadequate insights into their consumption patterns. Additionally, integrating renewable energy into the grid remains a challenge, and concerns about data privacy and security persist. Addressing these issues requires innovative solutions that enhance monitoring, reduce losses, and support a more efficient, secure, and sustainable energy infrastructure.

4. SYSTEM DESIGN

The system design for Smart Metering in Electricity Theft Detection and Power Monitoring is crafted to illustrate the interaction and flow of data between various components, utilizing system architecture, (DFDs), and Language (UML) diagrams. The system architecture provides an overarching view of core elements such as smart meters, data collection units (DCUs), a central server, and the communication network, detailing how data is collected, transmitted, and processed for both theft detection and power monitoring. DFDs break down these processes further by illustrating data sources, key functions, and storage points, while mapping the flow of data from the consumer's meter to the central server for analysis. To dive deeper into system interactions and behaviors, UML diagrams like use case, class, sequence, and activity diagrams offer specific insights into functionalities and processes. Together, these diagrams form a comprehensive framework for visualizing how the system efficiently detects electricity theft and enables robust power usage monitoring, ensuring clear data flow and organized control.



5. SYSTEM REQUIREMENTS

The system architecture includes three key components:

1. Distributor Side IoT Module:

- Measures the power sent from the distributor to the consumer.
- Contains sensors such as ZMPT 101B (voltage) and ACS 714 (current), interfaced with the ESP32 microcontroller.
- Sends data (voltage, current) to the server via Wi-Fi.

2. Consumer Side IoT Module:

- Measures the consumer's actual electricity usage.
- Compares data with the distributor-side readings to detect discrepancies (theft).
- Data is processed using the ESP32 and sent to the server.

3. API Server and Database:

- REST API built with Node.js and Express.js handles communication between the IoT modules and the mobile app.
- Data is stored in MongoDB, including sensor data and user details.
- Power theft is detected based on discrepancies in power data.

4. **Mobile Application:**

- Allows users to monitor power consumption in real-time.
- Displays alerts in case of electricity theft and offers historical data views.

SYSTEM IMPLEMENTATION PLAN

1. Phase 1: Requirement Analysis and Feasibility Study

- Gather detailed requirements from utility companies and consumers.
- Evaluate available hardware and software solutions, including sensors, microcontrollers, and cloud storage options.

2. Phase 2: Design

- Design the IoT modules (for both distributor and consumer sides), mobile app interfaces, and API server structure.
- Prepare circuit diagrams for the hardware and architectural design for the software stack.

3. Phase 3: Development

- **Hardware setup:** Assemble and program the ESP32 microcontrollers with voltage and current sensors.
- **Software development:** Build the API server, mobile app, and database. Begin implementing the core functionalities such as data collection, theft detection algorithms, and notifications.

4. Phase 4: Testing

- Perform unit testing on individual IoT modules, ensuring sensors accurately measure voltage and current.
- Integration testing for communication between IoT devices, API server, and mobile application.
- End-to-end testing, simulating electricity theft and monitoring system responses.

5. Phase 5: Deployment

- Set up the system in a controlled environment with actual users for beta testing.
- Deploy mobile app to Android and iOS platforms and the API server to a cloud service (e.g., AWS).

6. Phase 6: Maintenance and Updates

- Regular updates to improve theft detection algorithms.
- Ongoing monitoring of system performance and scalability, addressing hardware malfunctions, and software errors.

OTHER SPECIFICATIONS

ADVANTAGES

Here are the key advantages of smart metering for electricity theft detection and power monitoring:

- **Real-Time Monitoring:** Allows for continuous tracking of electricity consumption and immediate identification of irregularities.
- **Improved Theft Detection:** Advanced analytics help quickly detect unauthorized usage and tampering.
- **Cost Savings:** Reduces losses for utility companies by preventing theft, leading to more stable pricing for consumers.
- **Enhanced Accuracy:** Provides precise measurements of energy use, improving billing accuracy and customer trust.
- **Faster Response:** Enables quick action to address issues like power outages or theft incidents.
- **Energy Efficiency:** Helps consumers understand their energy use patterns, encouraging better energy management.
- **Data-Driven Decisions:** Utilities can optimize energy distribution and forecast demand based on detailed consumption data.

LIMITATIONS

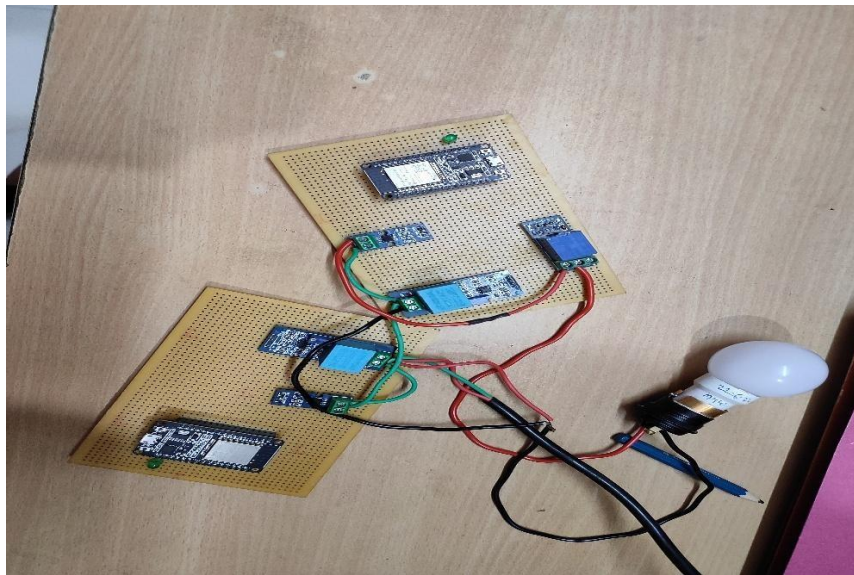
Here are some key limitations of smart metering for electricity theft detection and power monitoring:

- **High Implementation Costs:** Installing and maintaining smart meters and supporting infrastructure can be expensive.
- **Data Privacy Concerns:** Continuous data collection raises issues about user privacy and data security.
- **Technical Challenges:** Smart meters can face issues like software bugs, data inaccuracies, or network connectivity problems.
- **Vulnerability to Cyberattacks:** Smart meters and communication networks can be targets for hacking or cyberattacks.
- **Interference in Rural Areas:** Limited connectivity in remote areas can impact the effectiveness of data transmission.

7. RESULT AND DISCUSSION

Hardware Overview (As Seen in the Images)

- Your setup comprises: ESP32 Development Boards – Used for wireless communication, processing, and control.
- Relay Modules – To simulate load control (e.g., switching bulbs on/off).
- Current Sensors (ACS712 or similar) – To measure real-time current flow for energy monitoring.
- Breadboards or Perfboards – Neatly arranged for circuit connections.
- Connected Bulb – Acts as a test load for energy consumption.
- Laptop Interface (via USB) – Code running on Arduino IDE or VS Code to program the ESP32.



Working of the System

Smart Metering Functionality

- The current sensors continuously monitor the power consumption of the connected load (e.g., the bulb).
- These readings are sent to the ESP32, which calculates real-time power consumption.
- Data is displayed or transmitted (e.g., via Wi-Fi to a cloud/database/monitoring portal).

Theft Detection Mechanism

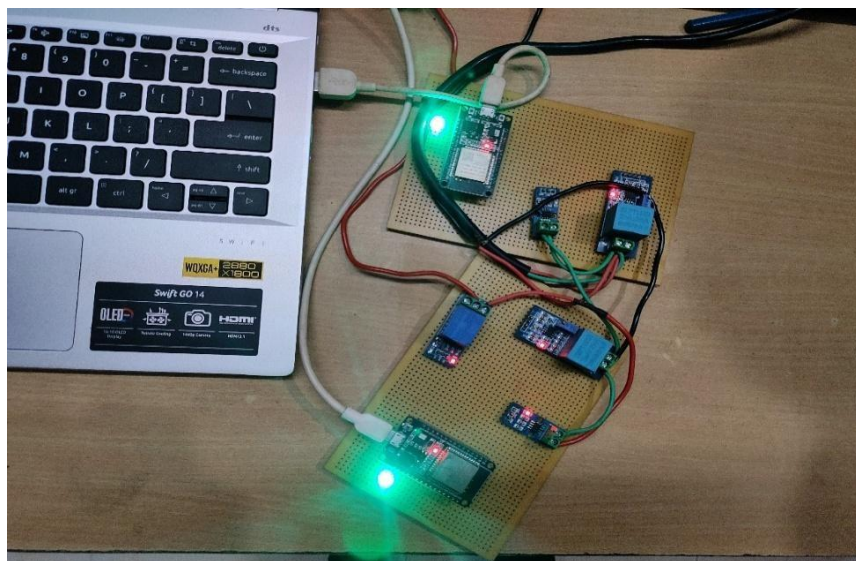
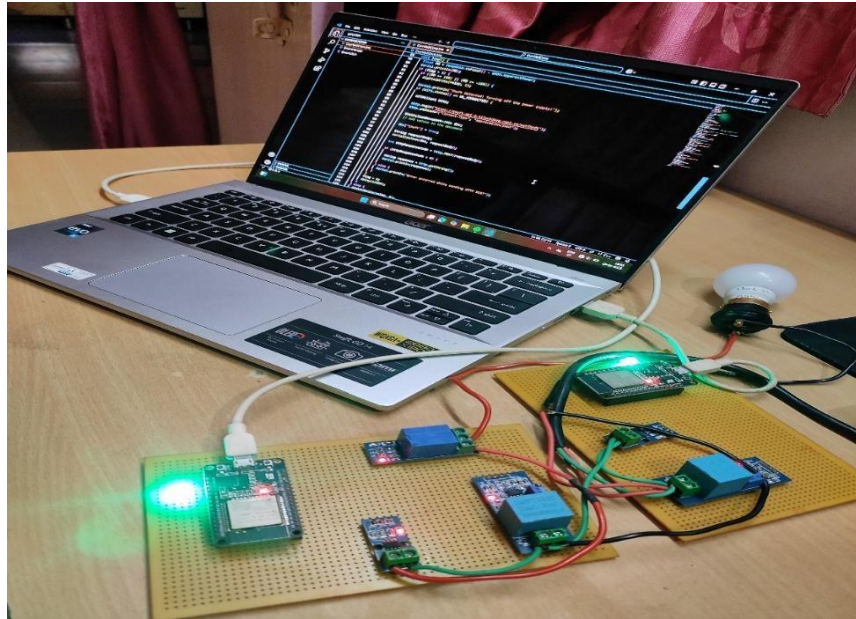
- The system uses two ESP32 modules representing two points: the main meter and the sub-meter (user-level monitoring).
- If power consumption recorded at the sub-meter is more than what's recorded at the main meter (due to illegal tapping), the system flags it as a potential theft.
- An alert can be triggered through a buzzer, LED indicator, or sent wirelessly to a web interface.

Relay Control

- The relays are used to simulate automated control based on usage patterns or in case of theft detection.
- For instance, in case of detected anomalies, the relay can disconnect the load to prevent misuse.

Monitoring and Control via Code

- As seen on the laptop screen, the ESP32 is running a script (likely in Python or Arduino C++) for:
 - Reading current values
 - Comparing values for theft detection
 - Controlling relays
 - Sending data to a serial monitor or IoT platform



8. CONCLUSION

The implementation of smart metering systems for electricity theft detection and power monitoring presents a significant advancement in ensuring efficient and reliable energy distribution. By integrating real-time monitoring, automated data analysis, and theft detection algorithms, this system not only helps in reducing power losses due to illegal consumption but also promotes transparency between energy providers and consumers.

The proposed system enhances the ability of utility providers to detect anomalies, monitor power usage patterns, and respond quickly to irregularities. It also empowers consumers with accurate and timely information about their electricity usage, encouraging more responsible consumption habits.

In conclusion, smart metering serves as a pivotal step toward modernizing the power infrastructure, improving operational efficiency, reducing revenue losses, and paving the way for a more sustainable and secure energy future.

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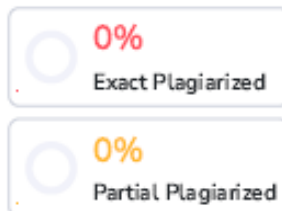
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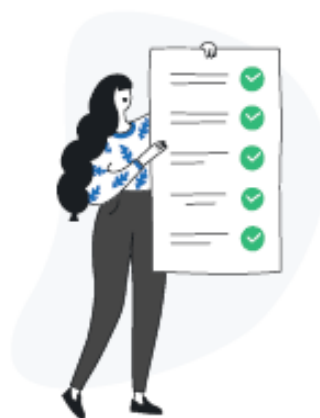
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